"UpaHanap: Enhanced Apartment Search And Listings With AR Technology In San Jose Del Monte"

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In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Computer Science

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ABSTRACT

Title of Research: **UpaHanap: Enhanced Apartment Search and Listings with AR Technology in San Jose del Monte**

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This study introduces UpaHanap, a mobile and web-based rental housing platform designed to address common challenges faced by tenants and landlords. By integrating property search, real-time communication, and secure payment features, UpaHanap streamlines the rental process and enhances user convenience.

Survey results revealed that 82.1% of tenants rely on online platforms, yet 65.8% encounter unreliable information and 55.3% find searches time-consuming. Meanwhile, 100% of landlords reported payment issues and emphasized the need for real-time updates and efficient communication tools.

UpaHanap provides verified listings, advanced search filters, and a unified interface for managing inquiries and payments. By bridging the gap between tenant needs and landlord requirements, the platform promotes reliability, transparency, and a smoother rental experience for all users.

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INTRODUCTION

The rental housing market presents significant challenges for both tenants and landlords. Tenants often face difficulties in finding rental properties that meet their specific needs, such as affordability, proximity to work or school, and essential amenities. The search process can be time-consuming and frustrating due to scattered and unreliable listings, limited availability in preferred areas, and challenges in communicating with landlords. These issues often leave tenants feeling overwhelmed and dissatisfied with their options.

On the other hand, landlords face their own set of hurdles in managing rental properties. High vacancy rates, late payments, and difficulties in finding tenants can disrupt their operations and financial stability. The lack of effective tools for advertising, communication, and property management further complicates their ability to meet tenant expectations and maintain consistent income. “One of the biggest challenges landlords face is attracting reliable, long-term tenants. The process of marketing the property, screening potential renters, and conducting thorough background checks is time-consuming. Many landlords underestimate the importance of vetting tenants, which can lead to issues such as late payments, property damage, or even legal disputes” (Rundy Huntley, Nov. 6, 2024).

These shared challenges highlight the need for a comprehensive solution that bridges the gap between tenant preferences and landlord needs. A study by Kamal, Mohd, and Baharun (2020) emphasizes that factors such as location, cost, property quality, and the overall environment significantly influence a tenant’s decision. Addressing these factors can lead to more satisfied tenants, reduced vacancy rates, and fewer operational challenges for landlords.

To address these issues, "UpaHanap" was conceptualized as a platform designed to streamline the rental process for both tenants and landlords. It offers features such as real-time updates, personalized search filters, video tours, and an intuitive communication system. By connecting tenants with properties that meet their preferences and providing landlords with tools to manage their listings and tenants efficiently,

"UpaHanap" aims to create a seamless rental experience for both parties. With these features, the platform seeks to improve tenant satisfaction, enhance landlord operations, and contribute to a more efficient rental housing market.

Background of the problem

The demand for rental housing has steadily increased over the years due to various socio-economic factors, including urbanization, migration for work or education, and the rising costs of property ownership. As a result, the rental housing market has become a critical component of the real estate sector. However, navigating this market remains a challenge for both tenants and landlords, as revealed by survey findings.

For tenants, the process of finding a suitable rental property can be time-consuming and frustrating. The survey results conducted by the researchers using interviews and online surveys indicate that 82.1% or 64 out of 78 respondents of tenants currently rely on online property platforms or social media groups to search for rental properties, which highlights the growing dependence on digital tools. However, 65.8% or 51 out of 78 respondents expressed frustration with scattered or unreliable information, making it difficult to make informed decisions. Additionally, 55.3% 43 out of 78 respondents identified time-consuming searches as a key pain point, while 44.7% 35 out of 78 respondents reported difficulty in contacting landlords and lack of affordable options. Tenants prioritize affordability (69.2%) or 54 out of 78 respondents, location (79.5%) or 62 out of 78 respondents, proximity to work or school (48.7%) or 38 out of 78 respondents, and amenities (e.g, parking, wifi)(35.8%) or 28 out of 78 respondents when selecting a property. Despite these clear preferences, 73.7% or 57 out of 78 respondents of tenants stated that limited availability in preferred areas remains a significant challenge. Furthermore, 61.5% or 48 out of 78 respondents of tenants are dissatisfied with the current options available for finding rental properties.

On the other hand, landlords face their own set of challenges in managing their rental properties. According to survey data, 100% of landlords encounter payment challenges with tenants, indicating the need for tools that ensure timely rent collection. Additionally,

100% of landlords consider real-time updates (e.g., availability status) for rental listings as very important, suggesting a strong demand for platforms that enable efficient property management. Communication also remains a key issue, with landlords currently handling tenant inquiries through in-person meetings (68.7%) or 54 out of 78 respondents and messaging apps (33.3%) or 26 out of 78 respondents. However, these methods often lack the structure and efficiency needed for seamless communication, especially when managing multiple tenants or properties. While various house-renting applications exist, they often fail to address these challenges comprehensively. The survey results show that tenants consider features such as verified property listings (61.5%), direct messaging with landlords (56.4%), user-friendly interfaces (46.2%), and advanced search filters (43.6%) as essential for rental platforms. Similarly, landlords expressed a unanimous interest in a centralized platform for managing tenant applications, payments, and communication, with 100% finding it "very useful.

Despite the existence of some rental platforms, their shortcomings—such as the lack of personalized search filters, ineffective communication channels, and fragmented payment systems—highlight the pressing need for a comprehensive, user-centric solution. This solution should bridge the gap between tenant preferences and landlord requirements while addressing the inefficiencies and frustrations in the current rental process. By integrating tenant and landlord needs into a single, efficient system, a new platform could revolutionize the rental housing market, fostering better experiences for all stakeholders.

This thesis aims to explore and propose a comprehensive solution that addresses these identified challenges, offering a more effective and seamless experience for tenants and landlords alike. The proposed solution will investigate how an integrated platform can address the key issues of finding apartments, unreliable information, payment challenges, and communication inefficiencies. Ultimately, the goal is to create an integrated solution that enhances the overall rental experience for all stakeholders involved.

General Problem

A notable gap in the rental housing market is the lack of landlord-focused applications that address the specific challenges landlords face in managing their properties. While there are several platforms for property listings and tenant searches, they tend to be more oriented toward tenants. This creates a situation where landlords often lack efficient tools to manage their operations. As a result, tenants may experience difficulties such as outdated property listings, limited communication options with landlords, and incomplete property details, which can contribute to frustration and missed opportunities.

Specific Problem

- **Lack of Communication Tools**

  Miscommunication between landlords and tenants often leads to delays in resolving issues such as maintenance requests or lease negotiations.

- **Inefficient Maintenance Request Handling**

  Maintenance concerns are often addressed informally, causing delays in issue resolution and dissatisfaction for both landlords and tenants.

- **Lack of Automated Legal Documentation**

  Landlords and tenants often face delays and inconsistencies when drafting lease agreements. Without a streamlined system, agreements may lack clarity or important details, leading to potential disputes and dissatisfaction.

- **Fragmented Property Handling Systems**

  Landlords managing multiple properties rely on manual methods or fragmented platforms, making it difficult to track listings, tenant inquiries, and property availability.

- **Tenant Dissatisfaction Due to Lack of Accessible Information**

  Tenants frequently experience frustration when they cannot easily access comprehensive and detailed information about a rental property.

- **Difficulty in Finding the Location of Rental Properties**

  Potential tenants often face challenges locating rental properties due to unclear or insufficient location information. Without detailed maps or proper directions, tenants may struggle to find the apartment or property, leading to frustration and wasted time. This lack of accessible location details can result in missed opportunities for landlords and inconvenience for prospective tenants.

- **Delayed Payments and Limited Payment Options**

  Tenants sometimes fail to pay on time, due to lack of payment options, causing financial instability for landlords.

Overview of the current state of the technology

Apartment Management Systems (AMS) simplify rental property management by automating tasks like property listings, tenant communication, rent collection, and maintenance requests. These platforms improve operational efficiency with features such as automated listings, integration with marketplaces like Zillow, in-app messaging, and online rent payment portals. Despite their benefits, current AMS solutions often fail to meet the specific needs of certain types of rental markets. This thesis explores these limitations and proposes a user-focused AMS design to address diverse landlord and tenant requirements.

Objectives of the Study

General Objective

The objective of this study is to develop a platform that enhances property management for landlords and simplifies the rental search for tenants. The platform will provide landlords with efficient tools for managing their properties, while offering tenants a seamless experience for browsing listings, accessing key property details, and navigating rental options with ease.

Specific Objectives

- **To Facilitate Communication**

  To develop an in-app messaging system that enables direct and secure communication between landlords and tenants, ensuring timely response and reducing misunderstanding.

- **To Develop Maintenance Request System**

  To create a formalized system for tenants to submit maintenance issues with detailed description and media, and for landlords to track and resolve these requests efficiently

- **To Develop a Customizable Lease Agreement Contract**

  To create a feature within the platform that generates a customizable lease agreement template, allowing landlords and tenants to collaboratively fill out essential terms and clauses in a structured and user-friendly manner. This ensures transparency, accountability, and legal compliance for both parties.

- **To Streamline Listing Management**

  To enable landlords to manage property details, availability updates and inquiries in real time.

- **To Develop an Augmented Reality-Based Tour**

  Implement a basic augmented reality feature that allows tenants to virtually tour available properties, giving them a realistic preview of the space before visiting in person.

- **To Integrate an Apartment Location Map for Upahanap**

  This feature will integrate Google Maps directly into the UpaHanap app, providing an interactive map that displays the exact location of listed apartments. Tenants can interact with the map within the Upahanap platform, zooming in or out and panning to explore the area around the apartment. Users can also view directions to the property without leaving the app, ensuring a seamless experience. The map will help tenants easily navigate to the apartment location and plan their visit, all while staying within the Upahanap platform, where the main activity and interaction take place.
  Integrate Google Maps into the UpaHanap app to show the location of listed

apartments. Tenants can interact with the map, zoom in, pan around, and view directions to the property without leaving the app. This feature makes it easy for tenants to find the apartment and plan their visit directly within the platform.

- **To Integrate a Payment System**

  Tenants can conveniently pay their rent via e-wallets, ensuring a seamless and efficient payment process for both tenants and landlords.

Scope and Limitations of the Study

The UpaHanap platform is designed specifically for landlords managing rental properties. As well as tenants who are trying to find the most specific house that fits their needs and budget. The app includes the following features:

Scope

- **Advanced filtering**

  Users can refine their property search based on specific preferences such as price range, apartment type, and whether the space is furnished or unfurnished. These filters streamline the search process, helping tenants find properties that match their specific needs.

- **Video Tours**

  Landlords can upload detailed video tours of their properties, showcasing key areas such as bedrooms, living spaces, kitchens, and bathrooms. This feature allows potential tenants to explore the property remotely, saving time and

providing a virtual walkthrough experience to help them make informed decisions.

- **Maintenance Request System**

  Tenants can easily report maintenance issues directly through the app, providing:

  - *Detailed descriptions*
  - *Photos or videos of the problem*

  Landlords can efficiently manage these requests by:

  - *Coordinating with maintenance personnel*
  - *Setting resolution deadlines*

    Notifications keep both parties informed about request updates, ensuring transparency and timely resolution.

- **In-app Messaging**

  Secure, direct communication between landlords and tenants. This will enable quick response to inquiries, and further improve discussion about rental terms, maintenance, or concerns.

- **Augmented Reality Tours**

  Tenants can explore properties remotely using AR, offering a virtual, 360-degree walkthrough experience. This feature is particularly useful for individuals who may be located in different cities or are unable to visit properties in person.

  Tenants can remotely explore properties using basic Augmented Reality (AR), viewing the inside of the unit by panning through a 360-degree virtual walkthrough. Additionally, tenants can navigate between different areas of the unit by clicking on directional arrows. This feature is particularly useful for individuals located in different cities or those unable to visit properties in person.

- **Notifications and Reminders via SMS and In-app Mail**

  A feature designed to keep tenants and landlords informed about important updates and tasks. Notifications include reminders for:

  - *Due dates for monthly rental payments or utility charges*
  - *Updates on the status of maintenance requests*

  This ensures timely communication and helps both parties stay organized and aware of their responsibilities.

- **Interactive Apartment Location Map**

  The map will display the location of apartments registered on the platform, allowing potential tenants to easily view and navigate to these properties. The map will show the general location of the apartment, providing directions to the property.

- **Payment System**

  Once tenants receive a notification about their monthly rental dues, they can conveniently make payments using an e-wallet through the platform. This streamlined process ensures quick, secure, and hassle-free rent transactions for both tenants and landlords.

Limitations

- **Reliance on User-Provided Content**

  The quality and accuracy of property listings and video tours depend on landlords' effort and honesty, which may vary.

- **Maintenance Tracking Challenges**

  While the platform provides a maintenance request system, the actual resolution

depends on landlords and personnel. Delays in completing requests could still lead to tenant dissatisfaction.

Local literature/studies/systems

Several studies have explored the development and use of mobile applications and technology-based platforms in addressing housing challenges. These platforms aim to improve the availability and accessibility of housing options, enhance convenience, and ensure the safety of tenants. These platforms seek to create more efficient and secure housing solutions for individuals facing difficulties in finding suitable accommodation.

Review of Related Literature

According to **Richestph** in their study about **Eviction in the Philippines: Challenges and Solutions for Landlords,** "The eviction process can be time-consuming, often stretching for months or even years. This delay may be caused by court backlogs, tenant appeals, and other procedural requirements. Many landlords find themselves unable to collect rental income during this period, exacerbating their financial burdens."

UpaHanap addresses these issues by including an automated notification system to remind tenants of upcoming due dates for rent payments, minimizing the risk of delayed payments that could otherwise lead to eviction proceedings.

As stated by **Respicio.ph**, landlords in the Philippines face significant struggles due to delayed payments, which affect their financial stability and overall operations. Delayed payments disrupt cash flow, making it challenging for landlords to meet property-related expenses such as mortgages, taxes, and maintenance costs. Notably, delayed payments often necessitate additional administrative efforts to track overdue accounts and implement late fee policies.

Moreover, **Richestph.com** emphasizes that persistent delays in rent payments frequently lead to legal complications. Landlords may need to serve notices or initiate eviction proceedings, which are both time-consuming and costly. This strain often results in heightened tension between landlords and tenants, making effective communication and clear lease agreements critical to resolving disputes.

The challenges surrounding delayed payments directly align with the objectives of systems like UpaHanap, which aims to bridge the communication gap and ensure transparent transaction processes. By integrating timely payment reminders and automated systems for tracking rental dues, such platforms offer landlords practical solutions to mitigate these issues and streamline their operations.

In the words of Mamurovet al. (2020) and Žalėnienė & Pereira (2021) on Student Satisfaction on Housing Services: Basis of Management Plan in the New Normal. It is one of the important functions of an institution of learning to arrange the students' living environment to be fully aware of their living conditions.

Stress that being well adjusted fits into the environment of which he is a part with an identity he has developed. Says Schmader & Sedikides (2018).

(Browning & Rigolon, 2019; Rodríguez-Hernández et al., 2020). also stated that student housing or residence has a substantial educational role. A definite relationship exists between the proper performance of academic functions and the place of dwelling of students.

Browning & Rigolon (2019) and Rodríguez-Hernández et al. (2020) argue that student housing significantly influences academic success, with a clear connection between the quality of housing and students' ability to perform academically. This is where UpaHanap fits into the conversation, as the app offers a platform that prioritizes the safe, comfortable, and well-located off-campus housing options. By considering factors like proximity to campus, safety features, and living conditions, UpaHanap helps students find accommodations that support their academic and personal development.

Review of Related Studies

As highlighted by Jacob Miller (2024) in the study Late Rent Payments Advice for Landlords: A Comprehensive Guide, late rent payments significantly disrupt landlords' 0cash flow, impeding their ability to meet essential property expenses such as mortgages, maintenance, and taxes. The study emphasizes that clear communication between landlords and tenants is crucial in mitigating these issues. Implementing legal measures, such as detailed lease agreements outlining payment terms and consequences for late payments, is also recommended to address persistent delays.

Furthermore, the research underscores the importance of establishing a structured payment system to ensure timely rent collection. By adopting proactive strategies, landlords can better manage financial challenges associated with delayed payments.

UpaHanap addresses these issues by implementing a payment reminder for both the tenant and landlord. With it, they can minimize the problem of having delayed or forgotten payments due.

In Malaga, J. R. B. (2022) study on Housing Facility and Accommodation of College Students: Inputs for Policy Development, home circumstances set conditions that influence the educational experience of students. The findings revealed that the majority of students were renting a housing facility outside the college. Mostly they were bed spacers staying in boarding houses that charged 1000 pesos or less as monthly rentals. The housing facilities they occupied were either made of concrete materials or mixed materials.

Students need to have a safe and comfortable environment where they are treated fairly and in a humane way. According to Percy (cited in Jackson et al., 2019), this directly connects to UpaHanap, an app designed to address the housing needs of students by providing a platform for accessible, safe, and well-located off-campus accommodations. Much like the study's findings, UpaHanap prioritizes affordable housing options,

allowing students to filter through various criteria such as price and type of housing (boarding houses, apartments, etc.). This way, students can make informed choices that align with their preferences and ensure a comfortable living environment conducive to both academic and personal success.

As described by Agosto, A., & Burdeos, A. (2022). on Macroeconomic Determinants of Housing Prices in the Philippines. The cointegration test result shows that GDP per capita is positively associated with housing prices by 1.24% in the long term, the inflation rate by 0.08%, and the unemployment rate by 0.398%. The major real estate boom in the Philippines happened in the periods 2000–2001, 2004–2005, and 2007–2008 before the financial crisis (PSA,2021). Conversely, in the same period, the interest rates, inflation rates, and unemployment rates declined (International Monetary Fund, 2019).

This study investigates the impact of macroeconomic factors on housing prices, influencing rental markets that cater to students. It provides insights into rental trends affecting students renting apartments and the challenges they face in finding affordable and accessible accommodations.

UpaHanap responds to these economic challenges by offering a platform that helps user navigate the rental market more effectively, even as macroeconomic conditions influence pricing trends. The app allows students to filter housing options by price, type, and location, making it easier for them to find suitable living spaces that fit their budget, even in times of rising housing costs.

Review of Related Systems

Rentpad: Search for apartments, condos, and houses for rent in the Philippines.

This is a popular platform in the Philippines that allows users to search for long-term and short-term rentals, including apartments, condos, and houses. It lists properties across major cities in the country, like Manila, Quezon City, and Cebu, and offers various filters such as price range and furnishing type.

Owning a property may be a goal for some Filipinos, but with the current state of our economy, it would be challenging to attempt to buy your own piece of real estate. For a lot of us, renting is the more practical choice, especially for millennials and Gen Zs. The majority of both generations value flexibility and freedom more than anything. Whether or not you are part of the demographic we have mentioned, we do want to showcase the advantages individuals have when they choose to rent their home. (Nina Tambal, Apr 08, 2023.)

UpaHanap also capitalizes on this trend by offering a digital platform that caters specifically to students seeking off-campus housing. Much like *Rentpad*, UpaHanap allows users to search for rental properties across various price ranges and locations, ensuring that users have access to a wide array of choices that match their needs and budget. The app's filters for price, amenities, and location align with the growing preference for flexible and affordable rental options, making it easier for students to find housing that fits their academic and personal lifestyle without the financial strain of ownership.

MyTown: Co-living apartments and dormitories for students and young professionals in the Philippines.

MyTown is a co-living space provider targeting young professionals and students in the Philippines, particularly those who work or study near the business districts. It offers co-living apartments and dormitory-style accommodations in areas like Makati and BGC. MyTown provides various amenities and an easy-to-use platform for booking. Given the absence of affordable accommodation in close proximity to major central business districts, young professionals did not have a choice but to commute for 3-4 hours daily in cramped public transportation options. business districts. It offers co-living apartments and dormitory-style accommodations in areas like Makati and BGC. MyTown provides various amenities and an easy-to-use platform for booking.

UpaHanap aligns with this trend by focusing on providing similar, affordable housing options for users or students, particularly those who need to be close to campus. By offering filters that prioritize proximity to campuses, amenities, and public transportation, *UpaHanap* mirrors the objectives of MyTown. The platform emphasizes affordability and convenience, addressing the same challenges of long commuting times and the need for cost-effective housing in central locations.

Hoppler: Connecting property seekers with brokers and property owners in the Philippines.

Hoppler is a real estate platform based in the Philippines that connects property seekers (both buyers and renters) with brokers and property owners. It offers a wide range of listings, including apartments, condominiums, and houses. The platform is focused on streamlining property transactions in the country.

“Our mission is to connect buyers and sellers of real estate with a network of people you can trust. Hoppler is the first fully online real estate network and brokerage company in the Philippines. We invite partner brokers and property owners to list their properties on our website and mobile app. Upon sourcing a potential buyer/renter, Hoppler assigns one (1) internal broker to represent the client.”

Like Hoppler's mission to connect clients with trusted brokers and property owners, UpaHanap can ensure that all listings are verified and curated to provide safe and reliable housing options for tenants.

Foreign literature/studies/systems

In the global context, rental housing challenges continue to grow as cities expand and demand for affordable accommodation outpaces supply. Various studies from international markets, particularly in developed countries, show that rising rental costs, urbanization, and economic inequality have intensified housing instability.

Review of Related Literature

As suggested by Perach, N., & Anily, S. (2022) on stable matching of student groups to dormitories. Apparently, not all properties of the single student's model continue to hold under group applications. Finally, we consider the incentive compatibility property of the proposed algorithm and describe a specific quasi-stable outcome for which no subset of student groups can gain by misrepresenting their preferences over the dormitories.

Housing is a major factor in determining the quality of life that students experience during their college/university years. Providing students with good on-campus housing often results in making higher education institutions more attractive for potential students and hence has been a major goal in the management of many of them (see Roche, Flanigan, & Copeland, 2010). Besides the room/apartment itself, a major factor that affects the quality of academic life of the students is the dormmates/roommates they are assigned with (see, for example, Araujo, Murray, 2010; Khan, Shekili, Badi, Khanbashi, 2020).

As observed by Jeff Rohde, in 10 things to do when a tenant consistently pays rent late, having a tenant consistently pay the rent late can be one of the most frustrating challenges a landlord faces. A tenant who always pays late can wreak havoc on budgeted cash flow and put a landlord in the situation of paying operating expenses late or having to go out of pocket until the monthly rent payment is received. Here are things landlords can do with tenants who consistently pay the rent late, along with tips to help landlords avoid ending up with tenants who don't pay rent on time. Tenants who consistently pay the rent

late may result in a landlord paying the property operating expenses late or paying with personal funds.

UpaHanap includes automated rent payment reminders to tenants, ensuring they are notified in advance of due dates. This proactive approach reduces the likelihood of missed or delayed payments.

As observed by Chris Lee in What Are the Best Practices for Dealing with Late Rent Payments? Late rent payments can severely disrupt a landlord's cash flow, forcing them to either delay property expenses or use personal funds to cover costs such as maintenance and mortgages. The article highlights that clearly defined rent payment policies and the use of online rent payment platforms can prevent late payments by streamlining the transaction process and ensuring transparency. Additionally, providing reminders and offering flexible payment methods are effective strategies to encourage tenants to pay on time.

UpaHanap addresses this issue by integrating features such as automated rent payment reminders, which notify tenants in advance of due dates. This proactive approach minimizes the chances of missed or delayed payments. The platform also incorporates an online payment system, making rent transactions more convenient and secure for both landlords and tenants, ultimately reducing financial strain and fostering better landlord-tenant relationships.

Review of Related Studies

As stated by Reedy and Company in 18 Acceptable Reasons for Late Rent Payments: A Landlord's Guide, late rent payments often arise from various tenant circumstances, including job loss, medical emergencies, and family crises. These situations can significantly disrupt landlords' cash flow, leading to challenges in covering operating

expenses like property maintenance and mortgage payments. The study emphasizes that understanding tenants' circumstances and offering flexible payment plans can help landlords maintain a positive relationship with tenants while ensuring timely payments.

UpaHanap addresses this issue by integrating automated rent payment reminders, ensuring tenants are notified in advance of due dates. Additionally, the platform offers a secure online payment system that provides flexibility and convenience, reducing the likelihood of delayed payments. By streamlining the rent collection process, UpaHanap helps landlords mitigate the financial strain caused by late payments while fostering better communication and understanding with tenants.

As observed by Azibo in Acceptable Reasons for Late Rent Payments, tenants may face various challenges that result in delayed rent payments, including job loss, medical emergencies, and banking errors. These circumstances can disrupt a landlord’s cash flow, leading to difficulties in covering essential property expenses such as maintenance and mortgage payments. The article highlights the importance of landlords understanding tenants’ situations and recommends offering flexible payment plans or grace periods to foster positive landlord-tenant relationships while ensuring financial stability.

UpaHanap addresses this issue by integrating features such as automated rent reminders and secure online payment options, reducing the likelihood of delayed payments. Additionally, the platform provides landlords with tools to monitor payment history and implement structured policies for late payments, ensuring that cash flow remains consistent and operational expenses are not adversely affected.

As observed by Booking Ninjas in Student Housing Lease-Up Period: The Time It Takes to Fill All Available Units, seasonal fluctuations in student housing often result in high vacancy rates during academic breaks. This transient nature of student populations can significantly impact landlords' occupancy rates, leading to lost income and potential financial strain. The article emphasizes the importance of marketing strategies and

offering flexible leasing options to reduce vacancies and maintain a steady cash flow during off-peak periods.

UpaHanap addresses this issue by offering features that help landlords manage seasonal vacancies. The platform includes tools for marketing properties to a wider audience, such as short-term tenants or professionals during academic breaks. Additionally, UpaHanap allows landlords to implement flexible leasing options and manage occupancy trends, ensuring higher occupancy rates and consistent rental income throughout the year.

Review of Related Systems

Zillow

Zillow's ecosystem of products and services is designed to provide a seamless, end-to-end transaction experience, helping renters, shoppers, buyers and sellers with every step of their journey. Founded to empower people with information so they could find the right home, Zillow is synonymous with real estate. As the most visited, most trusted name in the industry, "Zillow" is searched more often than "real estate,"2 and our data indicates 70% of all home buyers and sellers use Zillow at some point in their transaction.

UpaHanap can draw valuable insights from Zillow's approach to providing a seamless real estate experience, especially for renters. Zillow has built an ecosystem that supports renters through every step of their journey, from searching for apartments to completing the transaction. Similarly, UpaHanap could apply this model to help students streamline their apartment search process, ensuring they can easily find, filter, and connect with landlords offering accommodations near their campuses.

Renthop

A platform focused on New York City and other major US cities, RentHop ranks apartments based on their "hop score," considering factors like price and apartment quality. “We want to simplify your apartment search. Apartment hunting can be overwhelming, but we realized that finding a new home isn't about looking at every apartment listing, it's about looking at the best ones. Quality, not quantity.With the powers of math and technology combined, we created RentHop, a site that sorts listings by quality. We know you don't want to see listings without photos or from unresponsive landlords and brokers, so unlike other sites that only sort by price or posting date, our HopScore (a sophisticated set of algorithms) ranks apartment listings by quality, so only the best rise to the top.We currently serve most major U.S. cities.Scholarship and Internship Opportunity”

UpaHanap could take inspiration from RentHop’s innovative approach to apartment listings, particularly its use of the "HopScore" to rank listings based on quality rather than just price or posting date. Like RentHop, UpaHanap could develop a system that prioritizes apartments and housing options near campuses based on quality factors such as photos, landlord responsiveness, amenities, and safety features.

Realtor.com

A real estate platform that also includes rental listings. It provides detailed descriptions, photos, and tools for filtering by various criteria.

“For years, millions of home shoppers have turned to realtor.com® to find their dream home. Operated by Move, Inc., realtor.com® offers a comprehensive list of for-sale properties, as well as the information and tools to make informed real estate decisions. Today, more than ever, realtor.com® is The Home of Home Search℠.Realtor.com® also offers homeowners useful tools and resources through the My Home℠ dashboard. My

Home[SM] dashboard allows property owners to manage their home like the important investment it is by tracking their home's value over time, researching and managing home improvements, and scouting other similar properties in the neighborhood."

Similar to Realtor, UpaHanap also provides pictures and videos for students to view in order to specifically choose their home for their coming academic season.

SYNTHESIS

The reviewed literature and systems highlight the pressing challenges in housing, such as delayed rent payments, financial burdens on landlords, and the need for affordable and well-located accommodations for tenants. Existing platforms like Rentpad, MyTown, and international solutions like Zillow and RentHop provide valuable insights into addressing these issues through user-friendly interfaces, automated reminders, and quality-focused property listings. Similarly, studies underscore the connection between housing quality and academic success, emphasizing the importance of safe and affordable options for students. UpaHanap aligns with these findings by integrating features like automated notifications, secure payment systems, and customizable filters to bridge communication gaps, streamline rent collection, and ensure accessible housing for its target audience.

Building on insights from existing systems and research, UpaHanap offers an innovative solution to housing challenges in the Philippines. By addressing the specific needs of landlords and tenants through tailored features, it aims to create a seamless, transparent, and efficient housing experience that enhances tenant satisfaction and landlord stability while contributing to the broader housing ecosystem.

METHODOLOGY

This research will use an online survey-based methodology to gather data for the **UpaHanap** project, focusing on understanding the preferences and challenges tenants and landlords face in the apartment searching process.

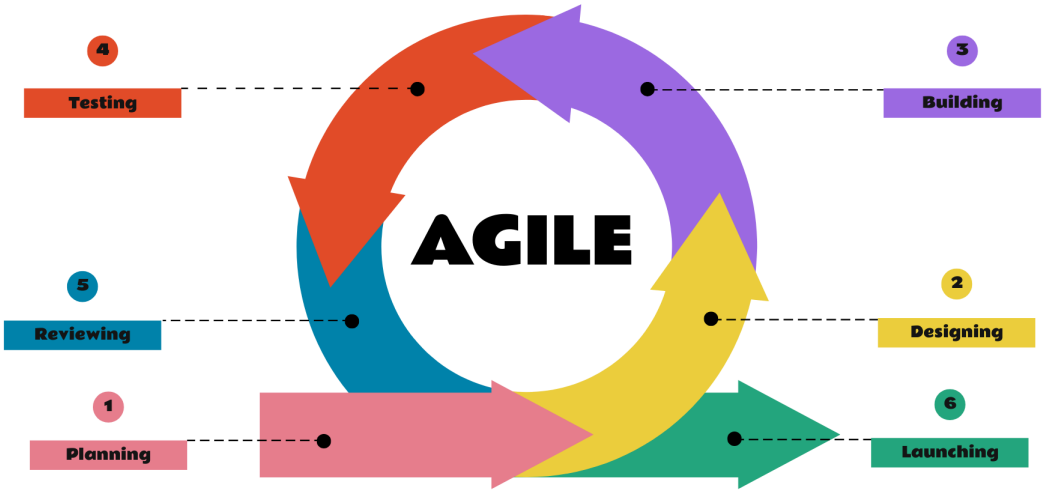


Figure 1. Agile Methodology

1. **Planning/Requirement gathering**
    - *Conduct surveys and interviews with users and landlords to understand their needs, preferences, and pain points in the apartment search and rental process.*
2. **Designing**
    - *Based on gathered information, design a user-friendly UI using existing platforms e.g. Figma.*
3. **Development**
    - *Use Flutter for cross-platform mobile app development to ensure consistency across Android devices.*

4. **Testing**
    - *Conduct a beta test of the application with tenants and landlords to assess usability, functionality, and effectiveness.*
    - *Gather user feedback for final improvements before launching the application.*
5. **Deployment**
    - *Deploy the application to target users and gather feedback for further improvement.*
6. **Review**
    - *Gather detailed feedback from early adopters regarding usability, performance, and bugs.*

Conceptual Framework



Figure 2. Input-Process-Output Model

The developer of the study used the input-process-output (IPO) model as the study's paradigm. The challenges faced by landlords, such as high property vacancy rates, difficulties in tenant communication, and the inefficiencies of managing multiple properties, serve as the input of the study. Additionally, the lack of streamlined tools for renters and landlords to efficiently connect and transact further highlights the need for this study. The software development methodology, which is the Agile Methodology, is what the researchers of the study used as a process to develop "UpaHanap: A Rental Property Management Platform for Landlords and Tenants." This process includes requirement gathering through surveys and interviews, designing an user-friendly interface with Figma, and developing a cross-platform mobile app using Flutter and Firebase. Testing and deployment phases ensure that the application is user-centric and functional. The study's output is UpaHanap, a mobile application that provides features such as advanced filtering options, in-app messaging, real-time updates, video tours, and maintenance request systems. The platform aims to reduce vacancy rates, enhance tenant-landlord communication, and streamline the management of rental properties.

System Design Specification

The UpaHanap system is designed to provide an organized, modern platform that connects renters (tenants) and property owners (landlords) with super-admin oversight for approvals and support. The overall design emphasizes usability, modularity, and efficient real-time data management through its primary modules: Developer, Super-Admin, Landlord, Tenant, and Support/Staff. The architecture follows a client–server model: the front end is built with Flutter (optimized for low-end Android and glassmorphism UI via UHScaffoldX, GlassCard, UHTheme), while the backend is powered by Firebase—handling authentication (Firebase Auth with phone/email OTP and role claims), database (Cloud Firestore for listings, bookings, verifications, support tickets), storage (Cloud Storage for images and documents), server logic (Cloud Functions for approvals, notifications, and webhooks), and push notifications (FCM). Optional integrations such as PayMongo (payments), Twilio/Vonage (SMS), Cloudinary (media pipeline), and Google Places (location/geo) extend the platform's capabilities without complicating the core client–server workflow.

Diagrams

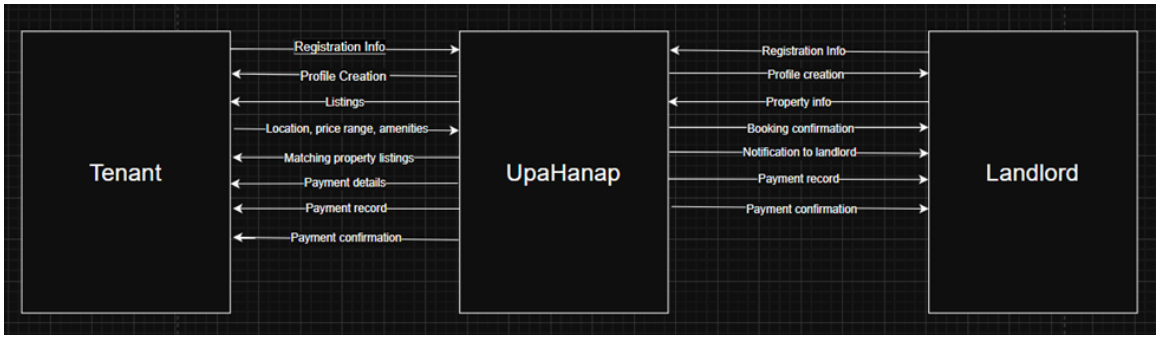


Figure 3. Data Flow Diagram Level 0

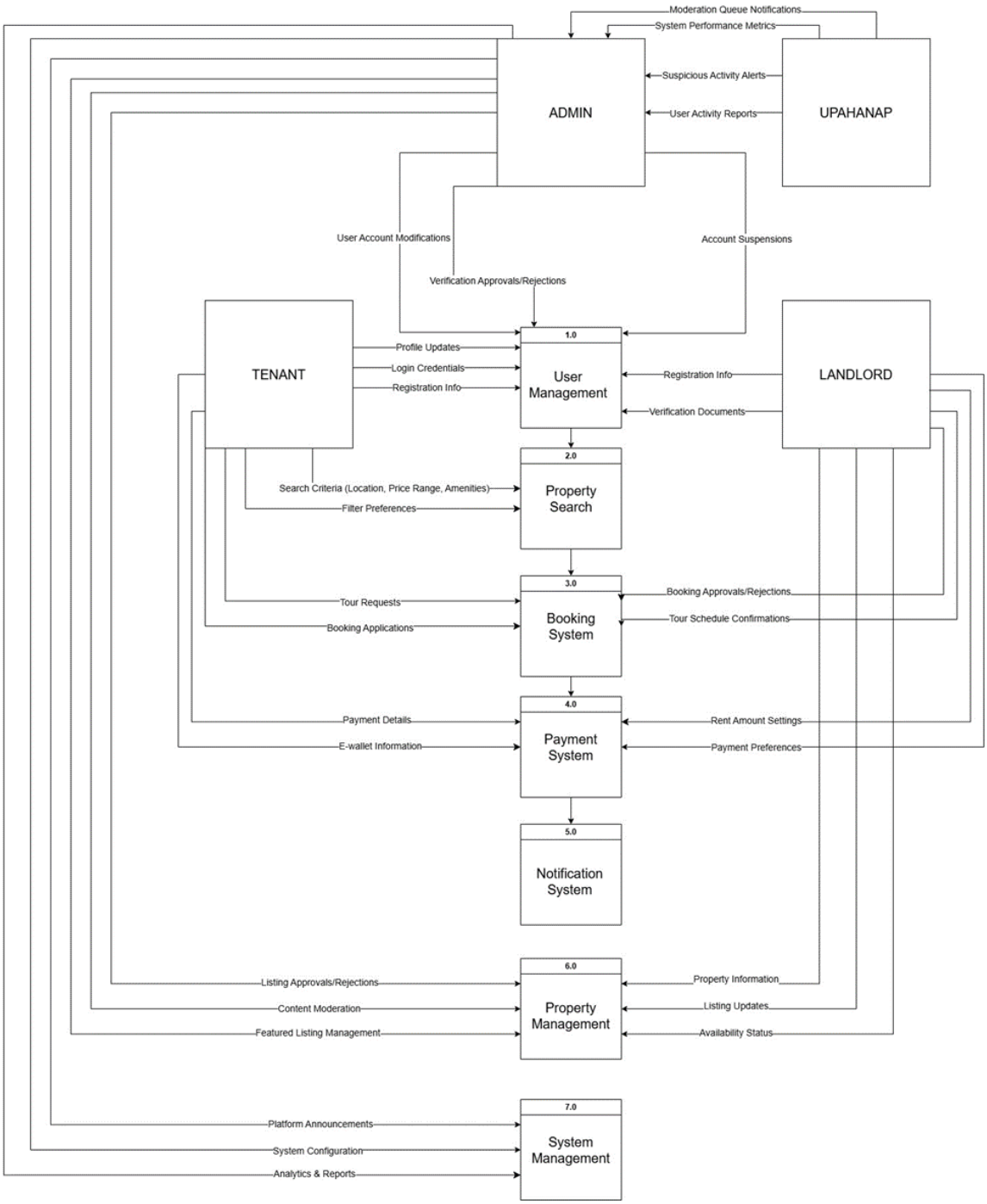


Figure 4. Data Flow Diagram Level 1

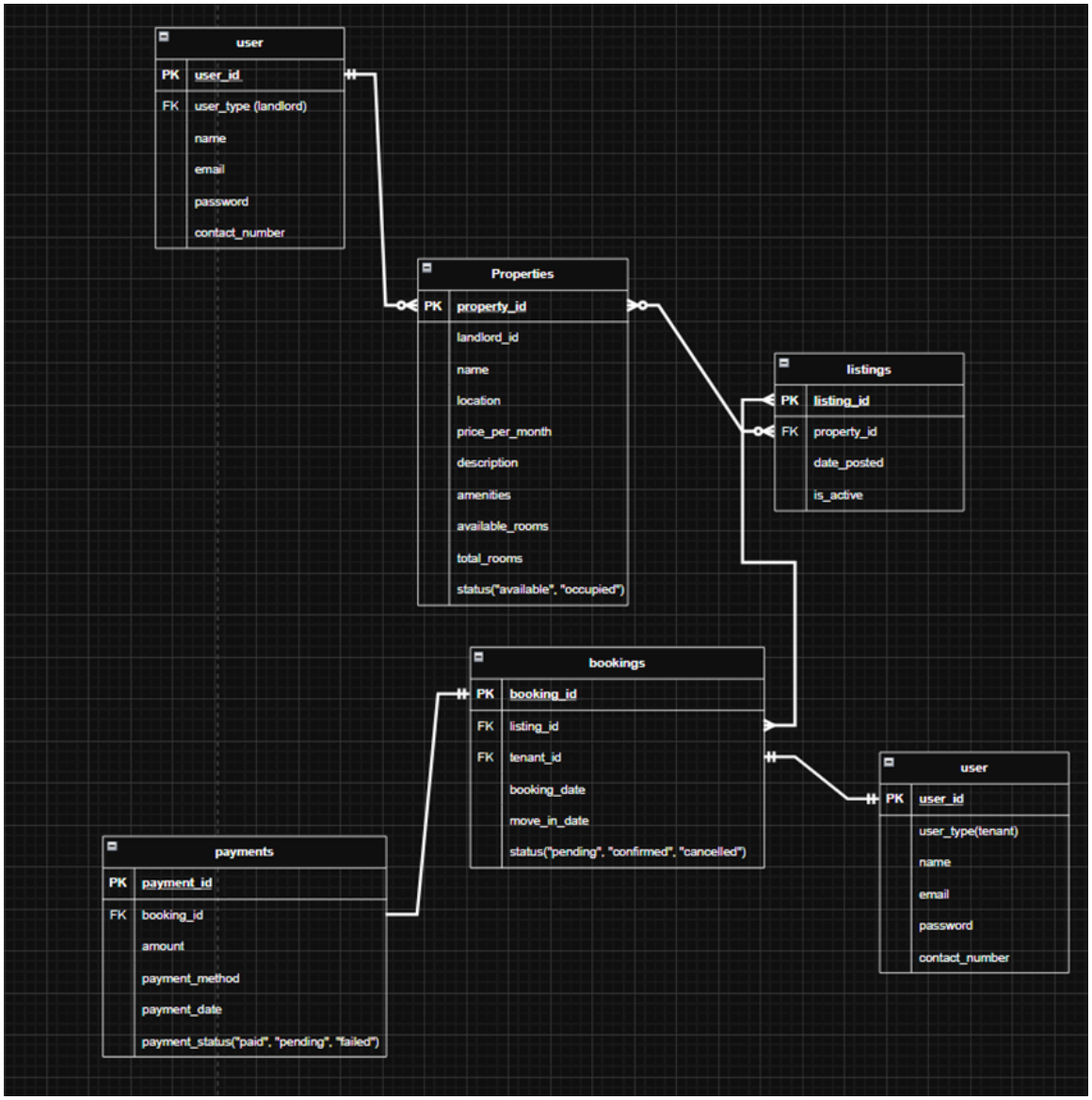


Figure 5. Entity Relationship Diagram

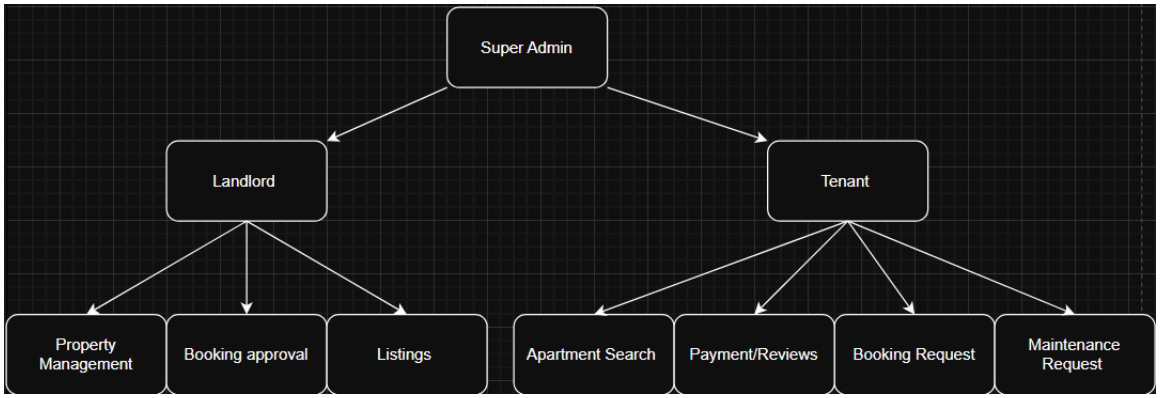


Figure 7. Hierarchical Chart

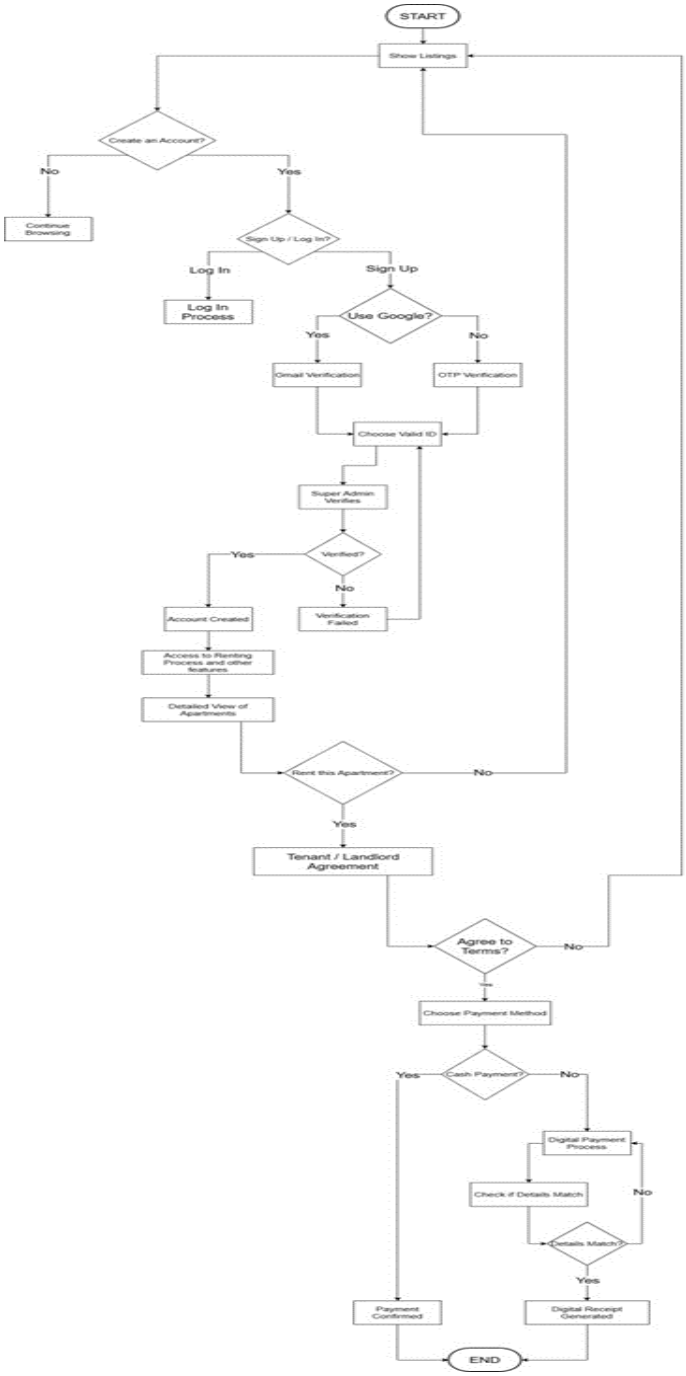


Figure 8.Landlord Flow Chart

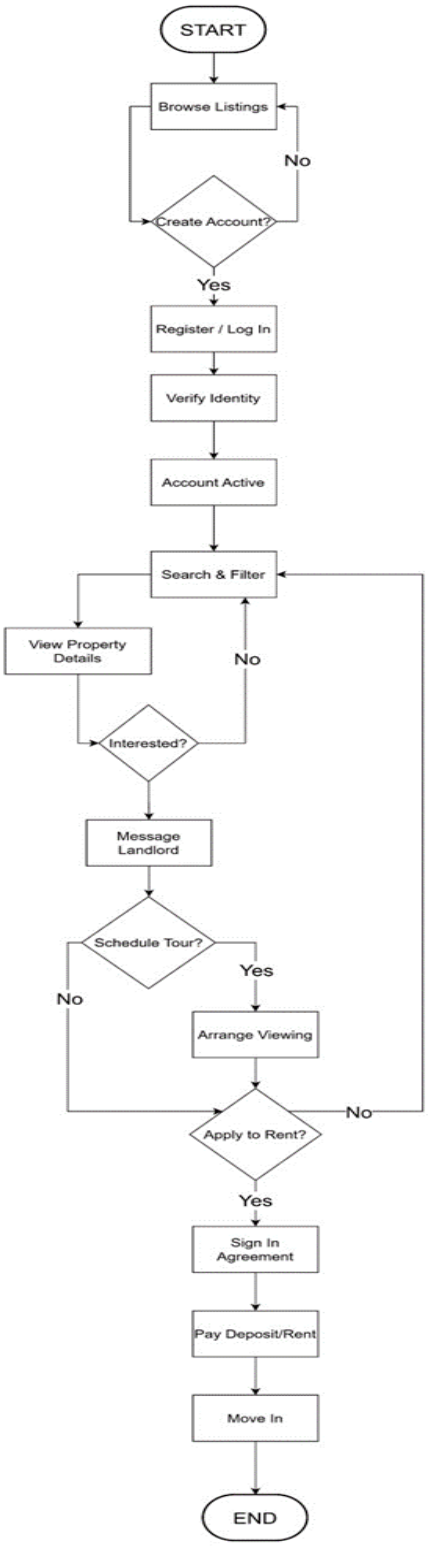


Figure 9. Tenant Flow Chart

Results and Discussions

Overview of the Results

A survey was conducted among **52 respondents** to assess the functionalities and overall user experience of the UpaHanap application in terms of effectiveness, usability, and reliability. This study aimed to evaluate how efficiently the system performs in helping tenants and landlords connect, manage listings, and handle rental transactions digitally. Specifically, the evaluation focused on the app's login and account access, property listing, search and filter functions, booking process, payment and verification systems, notifications and reviews, and overall design and navigation.

The responses gathered provided significant insights into how well the UpaHanap app meets its primary objectives—making property rental more accessible, convenient, and transparent for both tenants and landlords. The high level of agreement among participants indicates strong user satisfaction and confidence in the system's performance and ease of use.

Table 1. Mean Range Interpretation Legend

Mean Range	Interpretation
3.26 – 4.00	Very High (Strongly Agree)
2.51 – 3.25	High (Agree)
1.76 – 2.50	Low (Disagree)
1.00 – 1.75	Very Low (Strongly Disagree)

Data Gathering Procedure

The developers utilized the survey method as the primary data gathering procedure to obtain feedback from selected respondents regarding the usability and performance of the UpaHanap application. In choosing the participants, the purposive sampling technique was applied, focusing on individuals who have experience with digital rental platforms and mobile applications related to property listing. This ensured that the respondents could provide accurate insights relevant to the objectives of the study.

The survey was conducted through Google Forms, allowing respondents to evaluate various aspects of the UpaHanap app. Before answering, participants were informed of the purpose of the research, and a brief presentation of the app's features—such as login process, property listing, booking system, payment options, verification, and navigation—was provided. This ensured that each respondent fully understood the app's functionalities, resulting in reliable and accurate responses.

Statistical Treatment of Data

The developers employed the weighted mean as the statistical method to analyze the data gathered from respondents. This approach provides a more accurate representation of the respondents' level of satisfaction and agreement toward the different features of the UpaHanap application. By converting the collected data into a single mean score, it simplifies analysis and interpretation for decision-making and improvement planning.

The formula for the Weighted Mean is as follows:

Weighted Mean= $\frac{\sum(f \times x)}{N}$

Where:

- **f** = frequency of responses
- **x** = weight of each rating (1, 2, 3, 4)
- **N** = total number of respondents

User Feedback

Table 2. The login and account access process is easy and reliable.

Metric	4	3	2	1
Frequency	37	14	0	0
Weighted Mean	**3.73**			
Standard Deviation	**0.45**			

The results show a weighted mean of 3.73, interpreted as *Strongly Agree*. This indicates that users find the login and account access process easy, reliable, and efficient. The standard deviation of 0.45 suggests that responses were consistent, with most participants agreeing or strongly agreeing that account creation and authentication in UpaHanap are convenient and stable.

Table 3. The property listing feature makes it easy for landlords to post apartments and for tenants to view them.

Metric	4	3	2	1
Frequency	34	17	0	0
Weighted Mean	**3.67**			
Standard Deviation	**0.47**			

The property listing feature achieved a weighted mean of **3.67**, categorized as *Strongly Agree*. The result shows that the feature effectively supports both landlords and tenants by simplifying property posting and viewing. The standard deviation of **0.47** demonstrates a small variation in responses, confirming a high level of satisfaction and consistent usability of the feature.

Table 4. The search and filter options help users find the right property quickly.

Metric	4	3	2	1
Frequency	38	12	0	0
Weighted Mean	**3.76**			
Standard Deviation	**0.43**			

This feature obtained a weighted mean of 3.76, corresponding to a *Strongly Agree* interpretation. The search and filter function allows users to efficiently narrow down choices based on preferences such as location, price, and amenities. The standard deviation of 0.43 indicates uniform satisfaction, suggesting that users experienced minimal issues when searching for properties.

Table 5. The booking process is simple and updates both tenants and landlords properly.

Metric	4	3	2	1
Frequency	33	19	0	0
Weighted Mean	**3.64**			
Standard Deviation	**0.48**			

The booking process achieved a mean score of 3.64, with a standard deviation of 0.48. The data reveals that most users agree that the booking feature is straightforward and provides real-time updates between tenants and landlords. The small deviation signifies a consistent agreement, though a few participants suggested that further refinement in confirmation notifications could improve clarity.

Table 6. The payment options (GCash and Cash) make transactions convenient and easy to manage.

Metric	4	3	2	1
Frequency	35	16	0	0
Weighted Mean	**3.69**			
Standard Deviation	**0.46**			

This feature recorded a weighted mean of 3.69, interpreted as *Strongly Agree*. The result indicates that the inclusion of GCash and cash payments provides flexibility and convenience for both landlords and tenants. The low standard deviation of 0.46 shows stable and consistent positive responses, affirming that users appreciate the secure and accessible payment integration.

Table 7. The verification process (ID photo submission and face scanning) works smoothly and without difficulty.

Metric	4	3	2	1
Frequency	34	18	0	0
Weighted Mean	**3.65**			
Standard Deviation	**0.47**			

The verification process earned a mean of 3.65, which falls under *Strongly Agree*. This shows that respondents found the ID photo submission and facial verification process user-friendly and efficient. The standard deviation of 0.47 demonstrates that responses were generally consistent, validating the reliability of this feature in ensuring user authenticity and safety.

Table 8. The notifications and reviews help keep communication clear between tenants and landlords.

Metric	4	3	2	1
Frequency	43	9	0	0
Weighted Mean	**3.83**			
Standard Deviation	**0.38**			

The notifications and reviews feature achieved the highest mean of 3.83, interpreted as *Strongly Agree*. The findings indicate that this function significantly enhances communication between tenants and landlords. The low deviation of 0.38 reveals a strong consensus, proving that the notification system performs efficiently and helps maintain transparency and responsiveness within the platform.

Table 9. The app runs smoothly and loads quickly on my device.

Metric	4	3	2	1
Frequency	35	17	0	0
Weighted Mean	**3.67**			
Standard Deviation	**0.46**			

The performance evaluation of the application resulted in a weighted mean of 3.67 with an standard deviation of 0.46. This indicates that most users experienced smooth and stable app performance, with minimal lag or crashes. The consistency of responses confirms that UpaHanap is optimized for a wide range of devices, including low-end Android models.

Table 10. The app's design and navigation are user-friendly and easy to understand.

Metric	4	3	2	1
Frequency	40	11	1	0
Weighted Mean	**3.75**			
Standard Deviation	**0.47**			

The design and navigation feature obtained a mean score of 3.75, interpreted as *Strongly Agree*. The results demonstrate that respondents appreciate the simplicity, aesthetics, and intuitive layout of the UpaHanap interface. With a low standard deviation of 0.47, it indicates strong agreement and satisfaction regarding usability and design coherence across screens.

Table 11. Overall, the UpaHanap app provides a convenient and effective way to connect tenants and landlords.

Metric	4	3	2	1
Frequency	42	10	0	0
Weighted Mean	**3.81**			
Standard Deviation	**0.40**			

The overall effectiveness of UpaHanap recorded a weighted mean of 3.81, interpreted as *Strongly Agree*. This confirms that users perceive the app as a convenient and reliable bridge between tenants and landlords. The standard deviation of 0.40 suggests highly consistent responses, reflecting strong user approval and minimal variation in perception.

Expert Feedback

The software expert expressed a high level of satisfaction with the UpaHanap application, commending its modern, well-organized, and professional interface. They highlighted its clean and cohesive design, achieved through consistent color palettes, typography, and iconography that enhance readability and user engagement. The app's responsiveness across multiple Android devices was particularly praised, as it maintained stable performance and smooth transitions even on low-end hardware. The evaluator strongly agreed that the overall layout provides a seamless and intuitive experience that aligns with the platform's purpose of connecting tenants and landlords in a secure digital environment.

In terms of functionality, the experts unanimously agreed that all major modules of the application—such as login and role-based access control, property listing, booking and payment, and user verification—operated as intended. Each feature performed reliably throughout the testing phase without significant errors or system crashes. The role-based management system for tenants, landlords, and administrators was found to be secure and efficient, while the property listing feature accurately displayed uploaded details and images to tenants. The GCash and cash payment modules were verified to handle transactions securely and record data accurately, ensuring financial reliability. The verification process, which includes ID upload and face scanning, was also commended for its accuracy in confirming user authenticity.

In terms of reliability and security, the experts noted that UpaHanap demonstrated strong stability and data integrity throughout its evaluation. The system effectively handled simultaneous user interactions and maintained real-time synchronization of notifications and messages without delays. Its use of Firebase Authentication and Firestore Database ensured secure login validation, data protection, and controlled access to sensitive information. Furthermore, the database structure allowed accurate data storage and retrieval across all modules, confirming the system's efficiency in managing user and transaction data.

Under usability and efficiency, the evaluator found that the interface design promotes ease of navigation and consistent workflow. The glassmorphism-based layout, complemented by well-balanced spacing and icons, provides clarity and accessibility for users of all experience levels. The application performed smoothly, with minimal loading delays, maintaining responsiveness across different Android versions. The system's documentation and error-handling functions were also rated highly for maintainability, providing clear information for debugging and future updates.

Overall, the expert concluded that the UpaHanap application demonstrates strong usability, reliability, and functionality across all evaluation criteria. While a few minor refinements were suggested—such as optimizing tenant module spacing, improving notification margins, and adding pagination for apartment listings—the experts emphasized that these were minor design considerations rather than functional defects. They unanimously agreed that UpaHanap is a well-developed, secure, and user-centered

platform that meets the required software quality standards and is ready for real-world deployment.

Conclusion

The study aimed to evaluate the effectiveness, usability, and overall system quality of the UpaHanap application in addressing challenges related to property rental management within San Jose del Monte, Bulacan. A survey of fifty-two (52) respondents was conducted to assess the app's features, including login and account access, property listing, search and filtering, booking and payment, verification, notifications and reviews, performance, and interface design.

The findings revealed that the application consistently achieved "Very High" ratings across all major categories, with an overall weighted mean of 3.72 and an average standard deviation of 0.45, indicating strong user satisfaction and consistent responses. The login and account access feature was rated at 3.73, confirming that users found it reliable and secure. The property listing (3.67) and search and filter (3.76) functionalities were both found to be efficient and user-friendly, allowing users to locate and manage properties effortlessly. The booking (3.64) and payment (3.69) modules demonstrated smooth processing and transactional security, while the verification process (3.65) ensured accurate identity confirmation. The notifications and reviews feature received the highest weighted mean of 3.83, affirming its role in maintaining clear communication between users. The performance (3.67) and design and navigation (3.75) ratings likewise demonstrated strong user approval for app responsiveness and interface aesthetics.

The expert assessment validated these findings, confirming that UpaHanap effectively integrates its functional and visual components. The system's features were found to perform efficiently, securely, and accurately during evaluation. Both users and experts concluded that the app delivers a convenient and reliable digital platform that fulfills its intended purpose of connecting tenants and landlords. The high mean scores and minimal variance across results signify that UpaHanap achieved its development goals in terms of accessibility, stability, and usability.

Overall, the evaluation results affirm that UpaHanap is a stable, secure, and user-centered application capable of supporting digital rental management effectively. The app demonstrates high usability, visual consistency, and responsiveness across multiple devices, contributing to an enhanced user experience. Its strong backend reliability and security measures ensure accurate data handling and build user confidence. Although minor visual or layout improvements were identified, these do not significantly affect the app's performance or functionality. The system's quality standards and performance metrics confirm that it is ready for deployment in real-world usage.

Recommendations

Based on the results and conclusions of this study, several recommendations are proposed to further enhance the functionality, reliability, and overall user satisfaction of the UpaHanap application. The developers are encouraged to refine specific user interface elements by optimizing screen spacing within the tenant module and minimizing inner margins in the notification layout for improved visibility. Implementing pagination in the property listing section is also recommended to improve navigation and reduce excessive scrolling, allowing 10–15 listings per page for better content organization.

Additionally, the inclusion of a pop-up connectivity notification is advised to inform users when their internet connection is lost or unstable. This will enhance user awareness and reduce confusion during transactions or loading activities. The experts also suggested adding a reset button for the AR location feature, allowing users to quickly return to the default view for easier navigation.

Another important recommendation is to add an income analytics feature for landlords. This feature would allow landlords to view and track their earnings for each unit, or apartment. Providing simple summaries which can help them monitor their income and better understand their financial performance.

The development team is also encouraged to continue optimizing the application's performance to ensure smooth responsiveness across all Android devices, particularly low-end models. Maintaining consistent updates, system documentation, and bug tracking will further support the app's maintainability and long-term scalability.

These recommendations aim to improve the system's usability, efficiency, and accessibility, ensuring that the UpaHanap application continues to meet the evolving needs of both tenants and landlords. With its current level of performance, structure, and functionality, UpaHanap stands as a reliable, practical, and user-oriented digital rental platform ready for wide-scale implementation.

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APPENDIX A. CALENDAR OF ACTIVITIES AND GANTT CHART

Calendar of Activities

In the first week, the researchers began by assigning roles to ensure that every team member had a specific task, making the workflow efficient and organized. This week also involved brainstorming title proposals for the project, where multiple ideas were generated and discussed. The aim was to ensure the title reflected the goals and objectives of their study effectively. By the second week, after presenting their initial title to the panelists, the researchers realized it needed to be reassessed. The feedback from the panelists indicated a lack of clarity in the chosen title, prompting the team to refine their ideas and make necessary adjustments. In the third week, the researchers moved forward with requirements gathering. Surveys and interviews were conducted to collect user data and understand the specific needs of their target audience. This step was essential in ensuring the system's design and features were user-centered. Alongside this, they began drafting their research document, organizing data and insights collected so far. In

In October, the researchers focused on improving the system's features. In the first week, they evaluated which features were unnecessary and removed them, ensuring the system remained simple and user-friendly. By the second week, they began designing the UI/UX based on user feedback, creating wireframes that reflected the identified needs. In the third week, they conducted a thorough search for related literature, systems, and studies, ensuring their project was well-grounded in existing research.

In November, the team revised their paper based on expert advice and completed their Review of Related Literature by the second week. They also refined the system further and continued surveying users and landlords. The final weeks were dedicated to proofreading and finalizing their paper, correcting errors, and ensuring their system aligned with user needs. Every step contributed to the completion of a well-structured study

The Planning Phase, carried out from September to December 2024, involved designing the system architecture, defining user flows, and finalizing the technical requirements. During this period, the team created detailed wireframes and user interface prototypes using Figma, carefully mapping out both the tenant and landlord sides of the system. They also established the Firebase database structure that would handle user authentication, apartment listings, bookings, payments, and digital contracts. Security and data management strategies were reviewed to ensure privacy and reliability. By the end of December, the system blueprint, database schema, and overall project workflow were fully defined, marking the completion of the planning stage.

The Development Stage began in January 2025 and continued through November 2025. During January to March, the team focused on building the system's core functionalities. This included implementing Firebase Authentication with OTP verification, setting up

Firestore for real-time data handling, and creating the main navigation system for both tenant and landlord dashboards. The developers also built the apartment listing and browsing modules, ensuring that landlords could upload property information while tenants could explore listings dynamically through the app.

By April 2025, the project entered its next sprint, focusing on enhancing usability and integrating more advanced components. The team developed the Booking and Payment Modules, allowing tenants to book apartments and make payments through PayMongo's GCash API or select cash transactions. They also integrated a Messaging System, enabling communication between tenants and landlords once a verified rental agreement existed. Meanwhile, a Super Admin Panel was designed for overseeing user verification, report monitoring, and system activity tracking.

In May and June 2025, the team concentrated on improving user experience and responsiveness. They implemented the Submit Photo Screen and AR Upload Module, which allowed landlords to provide 360-degree apartment views, enhancing the realism of virtual tours. Interface animations and transitions were optimized for low-end Android devices to ensure smooth performance. Rigorous testing began, covering both mobile and web platforms.

From July to August 2025, the team conducted intensive debugging and feature refinement. Bugs encountered during testing were resolved, and redundant features were either simplified or removed. During this period, the developers also enhanced the Digital Contract Module, allowing tenants to view, review, and confirm lease agreements directly in the app. The Firebase Cloud Messaging (FCM) system was fully implemented, ensuring users received instant notifications for bookings, payments, and updates.

In September 2025, the project advanced to the Integration and System Optimization Phase. The Firebase backend was deployed to production, and live PayMongo credentials were tested successfully. The researchers ran full-scale tests using Android emulators and real devices to ensure all modules—authentication, payments, messaging, contracts, and AR view—functioned seamlessly. System logs were analyzed to fine-tune performance and reduce loading times.

By October 2025, the Production and Finalization Phase commenced. All modules were integrated and fully debugged. The UpaHanap Application now provided a complete experience, including tenant registration, property browsing, apartment booking, digital payments, contract signing, messaging, and notifications. The researchers then focused on final system validation, verifying that every feature aligned with the original project objectives.

Finally, in November 2025, the team concentrated on completing the research documentation and preparing for the final project defense. They revised all chapters, incorporated user feedback from the testing phase, and finalized thematic analyses and survey interpretations. Defense materials, such as presentation slides and demonstration videos, were also prepared. By the end of the month, the UpaHanap Application was finalized, demonstrating the team's successful transformation of conceptual plans into a

functional, user-friendly rental housing platform designed to simplify property search, management, and digital transactions for tenants and landlords alike.

Through these sequential phases, the developers of UpaHanap ensured that every stage of development—from planning and design to testing and deployment—was carefully executed, leading to a fully functional and user-friendly rental housing application ready for academic presentation and real-world implementation.

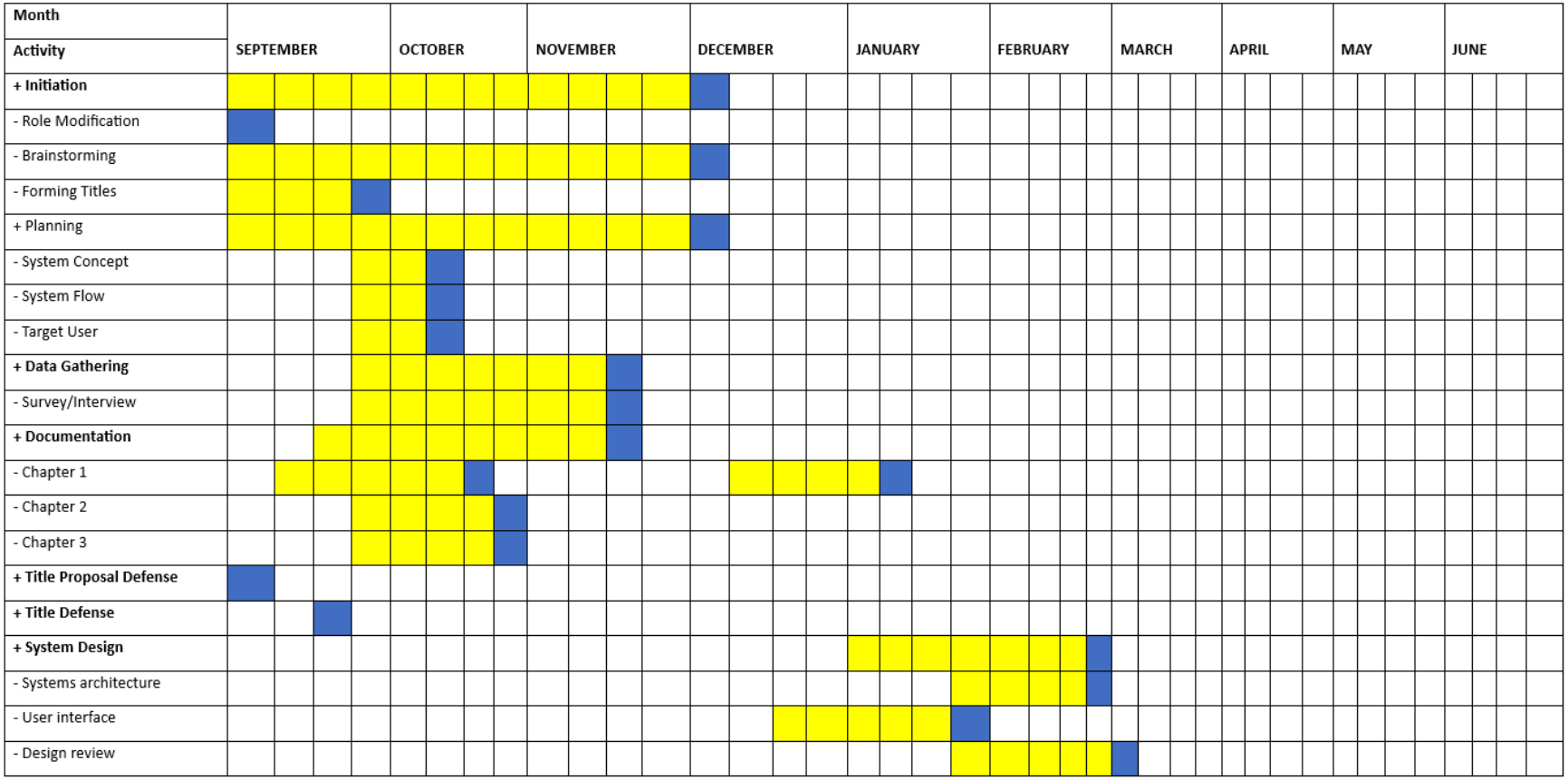


Table 12. Gantt Chart

Month / Activity	JULY				AUGUST				SEPTEMBER				OCTOBER				NOVEMBER			
- Chapter 4																				
- Chapter 5																				
+ Development and Iteration																				
- Implementation																				
- Coding																				
+ Testing																				
- Bug Hunting																				
+ Deployment																				
- System deployment																				
+ Review and Feedback																				
- User Feedback Collection																				
- Bug fixes																				
+ Defense																				
- Mock Defense																				
- Re-Defense																				

Table 12.1 Gantt Chart

APPENDIX B. ACTUAL THESIS EXPENSES

Thesis Expenses

Quantity	Specifics	Approximate Cost
1	Acer Nitro 5	₱34,000
1	Processor: Ryzen 7 5700x Ram: 16 gb GPU: Nvidia RTX 3060 12gb Storage: 512gb m.2 SSD	₱40,000
1	Insta360	₱25,000
1 month	Website Domain	₱100

Table 13. Actual Thesis Expenses

Prepared By:

John Mark D.C. Bautista

Jomari R. Pista

Nickos B. Poblete

Denver Mavric M. Rabino

Noted By:

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Irene M. Rabino

Approved by

Enoch James C. Tayanguna

Grace J. Orijuela, LPT

APPENDIX C. USER'S MANUAL

User Manual



Tenant Module

TENANT ONBOARDING SCREEN



Figure 10. Tenant Onboarding Screen

I. Layout Overview

Serves as the introductory screen that welcomes users to the platform before entering the Tenant module.

Displays the UpaHanap logo, tagline, and navigation indicators that guide users through the onboarding flow.

II. Functions

Logo and Tagline:

- Displays the UpaHanap logo prominently in the center.
- Tagline: *"The Shortcut to Your New Place"* — represents the platform's mission of making apartment searching quick and reliable.

Welcome Text:

- Greets users with *"Welcome to UpaHanap!"* followed by a short description: *"Find the perfect apartment easily."*
- Introduces new users to the app's purpose and core experience.

Navigation Buttons:

- **Skip:** Allows users to bypass the onboarding sequence and go directly to the Login or Tenant Landing Page.
- **Next:** Advances the user to the next onboarding slide in the introduction series.

Page Indicator (Dots):

- Shows the user's progress across multiple intro screens.
- The active dot is highlighted in blue to match the system's theme color.

Transition:

- Upon completion or skipping, users are redirected to the Tenant Login or Sign-Up Page to begin using the app.

TENANT LOGIN MODULE

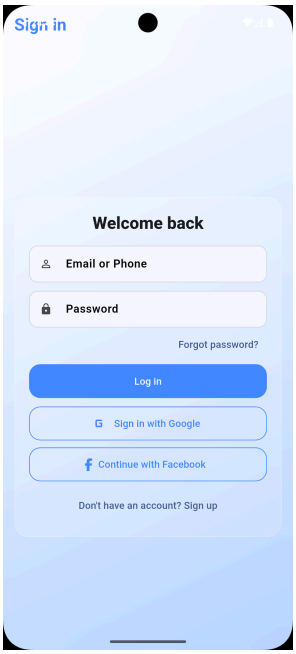


Figure 11. Tenant Login Page

I. Layout Overview

- Allows registered tenants to securely access their UpaHanap accounts.
- Provides access to personalized features such as saved apartments, booking history, and account settings.
- Includes input fields and quick links for account access and recovery.

II. Functions

- **Email / Phone Number:**
  - Enter registered login credentials for account access.

- **Password:**
  - Input the corresponding password for authentication.

- **Forgot Password:**
  - Opens a recovery process where tenants can reset their password using email or phone verification.

- **Login Button:**

- Verifies tenant credentials through Firebase Authentication.
- Redirects to the Tenant Home Page upon successful login.

TENANT HOME PAGE MODULE

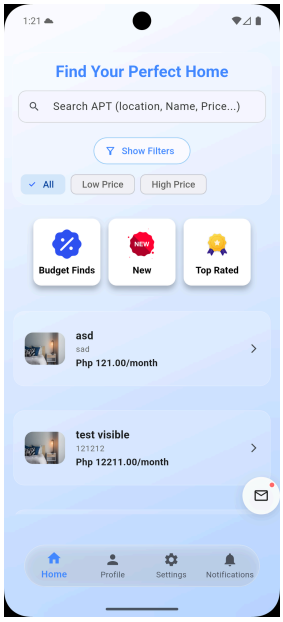


Figure 12. Tenant Home Page

TENANT LANDING PAGE

I. Layout Overview

This is the main entry interface for tenants upon launching the UpaHanap app.

It features a modern glassmorphic layout with soft gradients and animated visuals optimized for low-end Android devices.

The interface provides quick access to property browsing, filtering, and tenant account navigation through the bottom navigation bar.

II. Functions

Search Bar:

Allows tenants to search for apartments by location, name, or price.

Provides real-time Firestore integration for fetching apartment data dynamically.

Show Filters Button:

Expands a filter panel enabling users to refine listings by price range, amenities, and location.

Improves search precision and overall browsing efficiency.

Category Tabs (All / Low Price / High Price):

Filters displayed listings based on preset sorting logic (ascending or descending price).

Ensures users can quickly view budget-friendly or premium properties.

Shortcut Buttons (Budget Finds / New / Top Rated):

Displays specialized curated sections of apartment listings.

- **Budget Finds:** Lists affordable rentals below average market rate.
- **New:** Highlights recently added verified apartments.
- **Top Rated:** Shows highly rated apartments.

Apartment Listings:

- Thumbnail image
- Apartment title and brief description
- Monthly rental rate
- Navigation arrow linking to Book Now / Apartment Details Screen

Bottom Navigation Bar:

- **Home:** Returns to Tenant Landing Page
- **Profile:** Displays tenant profile, saved apartments, and preferences
- **Settings:** Provides access to app and account configurations
- **Notifications:** Shows announcements and updates
    - Alerts tenants about booking confirmations, landlord messages, and payment reminders.

APARTMENT DETAILS MODULE

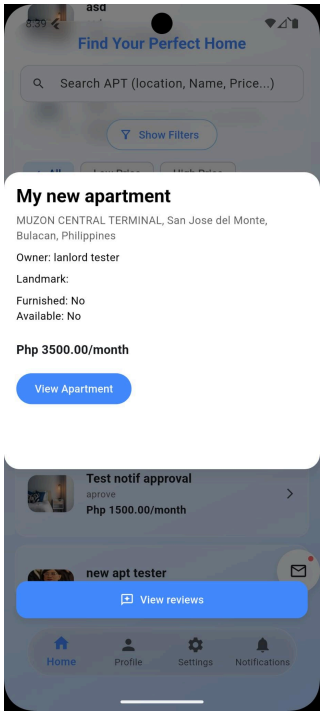


Figure 13. Apartment Details

I. Layout Overview

- Displays complete information about a selected apartment.
- Shows a carousel of apartment photos and videos at the top of the screen.
- Displays the apartment title, price, and location below the media section. Includes essential details such as amenities, unit size, and availability status.

II. Functions

- **Description:**
  - Provides a summary of the apartment's layout, style, and key features.

- **Amenities:**
  - Lists available facilities such as Wi-Fi, air conditioning, parking, and other inclusions.

- **Location Map:**
  - Displays the general area of the apartment for tenant reference.

- **Availability:**

- Indicates whether the apartment is currently open for booking.

- **Landlord Information:**
  - Shows the landlord's name, and contact option.
  - The Contact Landlord button allows tenants to send inquiries directly through the app.

- **Action Buttons:**
  - Save Apartment: Adds the property to the tenant's Saved List.
  - Book Now: Opens the booking screen for scheduling or reservation confirmation.
  - Back to Listings: Returns to the previous search results or the Home Page.

- **Photo and Video Viewer:**
  - Displays multiple photos and short videos uploaded by the landlord.
  - Tenants can swipe horizontally to browse all available media.
  - Videos provide a visual walkthrough of the apartment's layout and condition.

- **Augmented Reality (AR) Feature:**
  - A "View in AR" button is provided after the media section.
  - When tapped, the app opens a 360° interactive view of the apartment using AR technology.
  - Tenants can pan, rotate, and navigate between different areas of the unit using directional arrows.
  - If no 360° view is available, the system displays:
    "AR is not available for this listing."
  - This feature helps tenants visualize the apartment's space and layout before visiting in person.

SAVED APARTMENTS MODULE

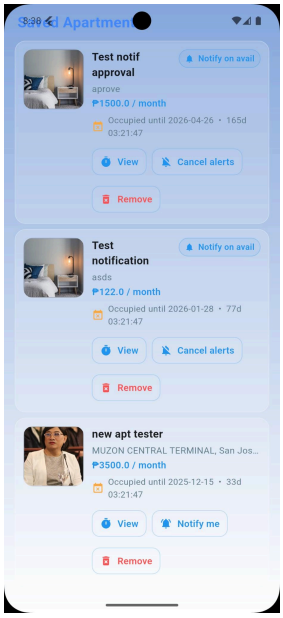


Figure 14. Saved Apartment Page

I. Layout Overview

- Displays a list of apartments that the tenant has marked as "Saved."
- Each card shows the apartment's photo, title, location, and price.
- A heart icon marks a saved apartment; tapping it again removes it.

II. Functions

- **View Apartment:** Opens the Apartment Details Module for the selected listing.
- **Book Now:** Opens the Booking Module directly from the saved apartment.
- **Remove from Saved:** Allows tenants to unsave a property instantly.
- **Sort & Filter:** Lets users organize saved apartments by price, location, or date added.

III. Sync and Storage

- Saved apartments are automatically stored in the tenant's Firestore account.
- Data syncs across all logged-in devices.

- Listings are private and accessible only to the tenant.

IV. Notifications

- Alerts appear if a saved apartment becomes unavailable or is booked by another user.
- The listing's status updates in real time to show availability changes.

BOOKING MODULE

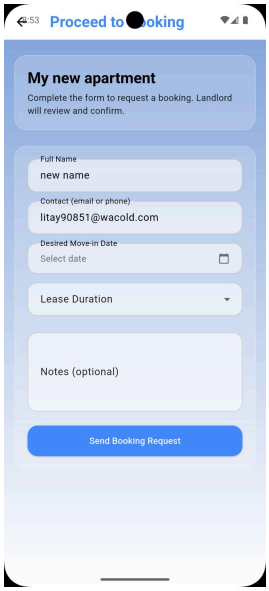


Figure 15. Booking Page

I. Purpose

- Allows tenants to reserve or schedule an apartment for rental or viewing.
- Ensures all bookings are properly recorded and confirmed through the system.

II. Booking Form

- Tenants fill out a simple booking form after selecting Book Now from the Apartment Details page.
- The form includes the following fields:
    - **Full Name** – automatically filled in from the tenant's profile.
    - **Contact Number** – used for booking confirmation.

- **Apartment Name** – displays the selected apartment's title.
- **Preferred Date** – date for visit or rental start.
- **Duration** – specifies the stay period or rental length.
- **Special Requests/Notes** – optional field for additional instructions or inquiries.

III. Booking Confirmation Process.

- A "Booking Successful" message appears once the request is sent.
- The landlord receives a notification about the new booking.
- The tenant can view the booking status (Pending, Approved, or Declined) in their Bookings Page.

IV. Payment or Approval Status

- **Pending:** Waiting for landlord approval.
- **Approved:** Booking has been accepted and confirmed by the landlord.
- **Declined:** The request was rejected; tenants may rebook or contact the landlord.
- If payment integration is enabled, tenants may be redirected to a secure payment gateway (e.g., PayMongo) after approval.

V. Notifications

- The tenant receives real-time notifications when the landlord updates the booking status.
- Push notifications appear through the app and the notification bell icon.

PAYMENT MODULE

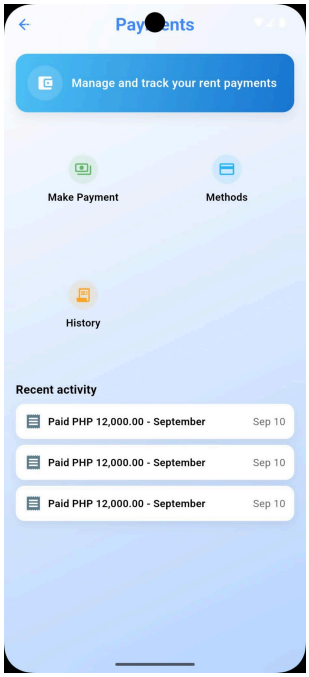


Figure 16. Payment Page

I. Layout Overview

- Allows tenants to complete rental or booking payments securely within the app.
- Activated once a booking has been approved by the landlord.
- Displays the total amount due, reference number, payment options, and transaction status.

II. Functions

- **View Payment Details:** Shows the total amount, breakdown (rent, deposit, fees), and payment deadline.
- **Select Payment Method:**
    - **GCash via PayMongo:**
        - Opens the integrated PayMongo payment page.
        - Tenants can enter the GCash number of recipient to pay directly through the secure gateway.

- The system automatically verifies the transaction upon completion.

- **Cash Basis:**
  - Tenants may choose to pay directly in cash to the landlord.
  - A confirmation form appears where tenants can mark the payment as "Cash" and upload an optional proof of payment (e.g., photo of receipt).
  - The landlord or admin will manually verify and confirm the cash transaction.

- **Upload Proof of Payment:** Allows tenants to attach a receipt or screenshot if payment is made outside the app.
- **Confirm Payment:** Submits payment confirmation for landlord or admin review.
- **View Payment History:** Displays all completed or pending transactions with timestamps and reference numbers.
- **Download Receipt:** Generates a digital receipt for completed payments.

III. Notifications

- Tenants receive a confirmation alert once the payment is successfully processed or verified.
- Notifications are sent if a payment is pending review, declined, or requires re-submission.
- Landlords and admins are automatically notified once a tenant confirms payment.

MESSAGING MODULE

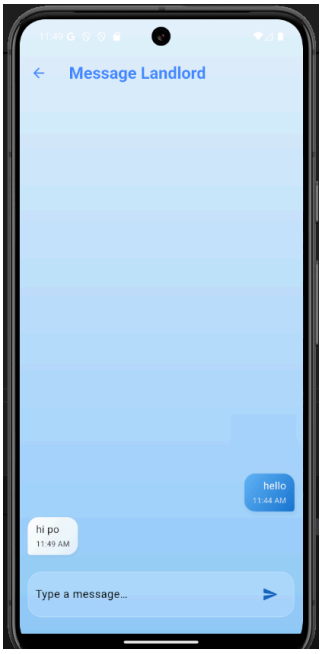


Figure 17. Messaging Page

I. Layout Overview

- Provides a built-in chat feature for tenants to communicate with their landlord regarding their rented apartment.
- Accessible only after a booking has been approved or confirmed.
- When no official apartment is assigned, the module is locked.

II. Functions

- **Open Chat:** Allows the tenant to view and continue existing conversations with their landlord.
- **Send Message:** Sends text updates or questions directly to the landlord.
- **View Conversation History:** Displays all past messages related to the apartment.
- **Back to Dashboard:** Returns the tenant to the main Home Page.

III. Notifications

- Tenants receive instant notifications when the landlord replies to a message.
- Shows time stamps for each message.

- If the apartment contract ends or the booking is cancelled, the messaging access automatically disables.

MAINTENANCE REQUEST MODULE

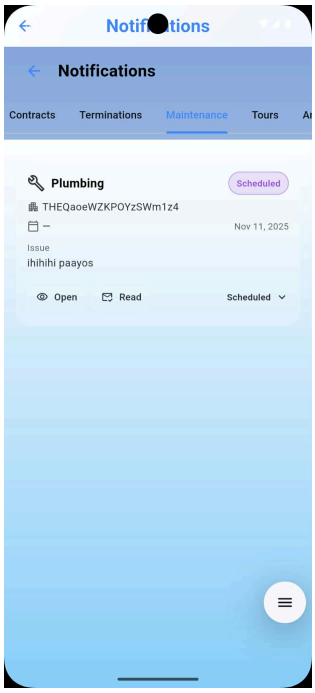


Figure 18. Maintenance Request Page

I. Layout Overview

- Allows tenants to report maintenance issues or repair needs directly to the landlord
- Includes fields for describing the problem, selecting the type of issue, and attaching a photo or video for reference.
- Displays a list of submitted requests with their current status.

II. Functions

- **New Request:** Opens a form where tenants can describe the maintenance issue.
- **Category Selection:** Lets users choose the type of problem (e.g., plumbing, electrical, appliance, structural).
- **Attach File:** Allows tenants to upload a photo or video showing the issue.
- **Submit Request:** Sends the report to the landlord or maintenance team for review.

- **View Status:** Displays the progress of the request (Pending, In Progress, Completed).

III. Notifications

- Tenants receive alerts when their request is acknowledged, updated, or completed.
- The system notifies both the landlord and tenant in real time when changes occur to the request status.

PROFILE MODULE

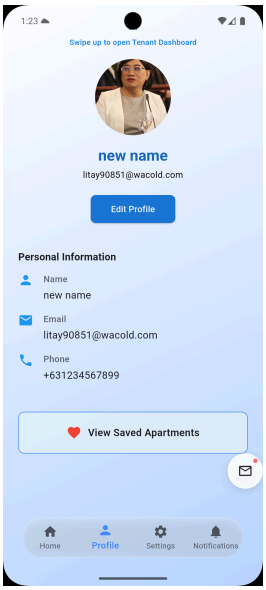


Figure 19. Tenant Profile

TENANT PROFILE MODULE

I. Layout Overview

Displays the tenant's personal information

II. Functions

Profile Photo:

- Displays the tenant's uploaded image in a circular frame.
- Tapping or editing allows users to upload or replace their profile picture using Firebase Storage.

Personal Information Section:

- Shows tenant's full name, email, and contact number.
- Data is dynamically fetched from Firestore and displayed in real time.

Edit Profile Button:

- Opens the Edit Profile Form where tenants can update their name, email, or contact number.
- Automatically syncs changes with Firestore upon saving.

View Saved Apartments:

- Directs the tenant to a dedicated page listing apartments they have marked as favorites.

Swipe Up Gesture:

- Displays an overlay transition or navigates to the Tenant Dashboard for advanced account management.

HELP & SUPPORT MODULE

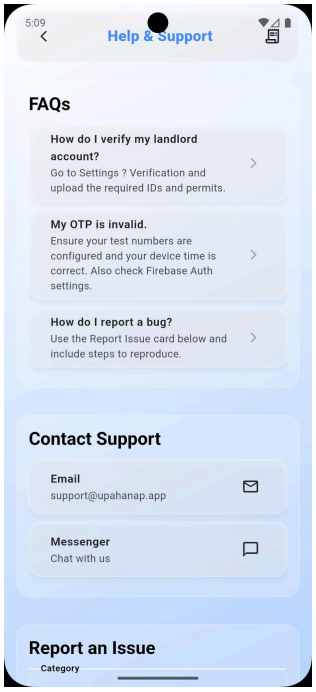


Figure 20. Help & Support Page

I. Layout Overview

- Provides tenants with access to assistance and support from the UpaHanap admin team.
- Contains options for submitting inquiries, reporting issues, and viewing previous support requests.
- Includes a message input box, attachment option, and a send button.

II. Functions

- **New Support Request:** Opens a form where tenants can describe their issue or concern.
- **Category Selection:** Allows users to select the type of concern (e.g., technical issue, account problem, booking concern).
- **Attach File:** Lets tenants upload screenshots or documents related to their concern.
- **Send Message:** Submits the request directly to the Super Admin for review.
- **View Response:** Displays replies or updates from the admin within the same conversation thread.

- **Close Ticket:** Marks a resolved concern as completed.

III. Notifications

- Tenants receive alerts when the Super Admin responds to their support request.
- The system updates the ticket's status in real time (Pending, In Progress, Resolved).

Landlord Module

LANDLORD ONBOARDING SCREEN



Figure 21. Landlord Onboarding Page

I. Layout Overview

Serves as the introductory screen that welcomes users to the platform.

Displays the UpaHanap logo, tagline, and navigation indicators that guide users through the onboarding flow.

II. Functions

Logo and Tagline:

- Displays the UpaHanap logo prominently in the center.
- Tagline: *"The Shortcut to Your New Place"* — represents the platform's mission of making apartment searching quick and reliable.

Welcome Text:

- Greets users with *"Welcome to UpaHanap!"* followed by a short description: *"Find the perfect apartment easily."*
- Introduces new users to the app's purpose and core experience.

Navigation Buttons:

- **Skip:** Allows users to bypass the onboarding sequence and go directly to the Login or Tenant Landing Page.
- **Next:** Advances the user to the next onboarding slide in the introduction series.

Page Indicator (Dots):

- Shows the user's progress across multiple intro screens.
- The active dot is highlighted in blue to match the system's theme color.

Transition:

- Upon completion or skipping, users are redirected to the Tenant Login or Sign-Up Page to begin using the app.

LOGIN MODULE

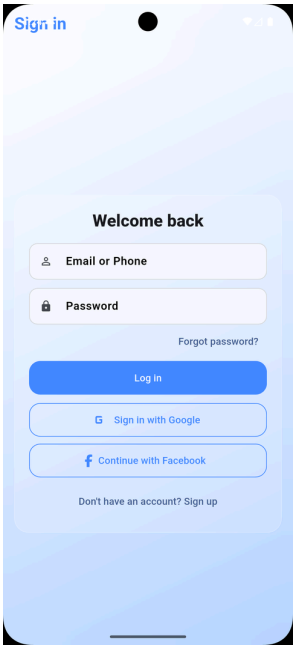


Figure 22. Login Page

Landlord Login Module

I. Login Interface

- Contains fields for Email/Phone Number and Password.
- Provides Forgot Password support for account recovery.
- Displays a Login button to initiate verification.

III. Authentication Process

- Credentials are verified through Firebase Authentication.
- Face Scanning for Verification confirms the landlord's identity securely and conveniently by matching their live facial scan against their stored image from verification.
- Verified landlords are redirected to their Landlord Dashboard.
- If verification is pending, the system prompts users to complete ID Verification first.

IV. Sign Up Redirect

- Includes a Sign Up link for new landlords who want to create an account.
- Redirects to the Landlord Registration Module for completion.

LANDLORD DASHBOARD MODULE

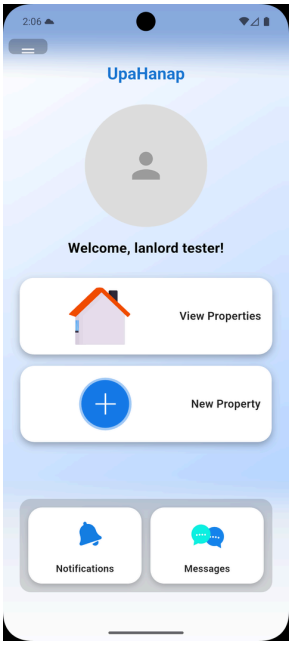


Figure 23. Landlord Dashboard

I. Layout Overview

- Displays the main control interface for landlords after logging in.
- Shows the landlord's name or email at the top, along with a profile icon.
- Contains primary buttons on the main screen: View Properties, New Property, Notifications, and Messages.
- A burger menu is located at the upper-left corner, giving access to all landlord features and system settings.

II. Functions

Main Dashboard Buttons

- **View Properties:**
    - Opens the list of all active property listings owned by the landlord.
    - Allows viewing, editing, or deleting existing apartments.

- **New Property:**

    - Redirects to the Add Property Module to create and upload new listings.

- **Notifications:**
    - Shows updates such as booking requests, tenant payments, or maintenance issues.

- **Messages:**
    - Opens the Messaging Module for communication with confirmed tenants.

Burger Menu Options

- **Dashboard:**
    - Returns to the main landlord overview with stats and shortcuts.

- **My Properties:**
    - Opens the full list of properties managed by the landlord.
    - Provides access to edit, update, or delete listings.

- **Add Property:**
    - Shortcut to add new apartment listings with details, images, and AR files.

- **Bookings:**
    - Displays tenant booking requests and scheduled rentals for review, approval, or decline.

- **Maintenance:**
    - Shows maintenance requests submitted by tenants.
    - Allows updating the request status (Pending, In Progress, Completed).

- **Messages / Support:**
    - Opens the chat center for tenant communication and admin assistance.

- **Payouts:**
    - Displays landlord earnings, transfer history, and payment details.

- **Settings:**
    - Opens account preferences, profile editing, and verification details.

- **Log Out:**
    - Signs the landlord out of their account and returns to the login page.

III. Notifications

- Alerts appear for new booking requests, maintenance issues, or tenant messages.
- Updates in real time whenever a tenant interacts with a landlord's listing or sends a payment confirmation.

ADD PROPERTY MODULE

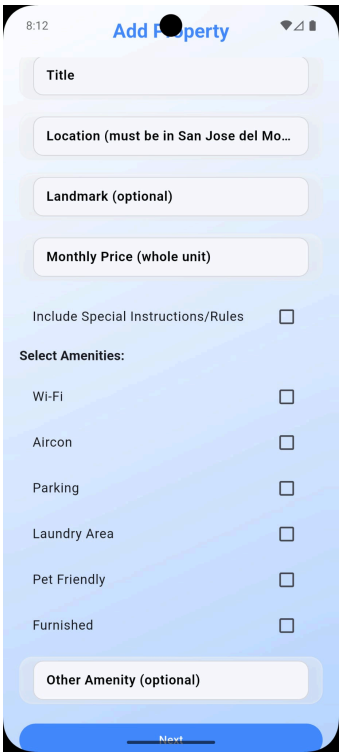


Figure 24. Add Property Page

I. Layout Overview

- Allows landlords to upload new apartment listings to the UpaHanap platform.
- Consists of three main sections: Property Details, Pin Apartment Location, and Submit Photos & 360° View.
- Each section must be completed before submission to ensure accuracy and a complete viewing experience for tenants.

II. Functions

1. Property Details

- **Title:**
    - Enter the name or label of the apartment

- **Location:**
    - Input the apartment's address or area for easy tenant visibility.

- **Landmark:**
    - Add a nearby landmark (e.g., "Beside City Mall" or "Near MRT Station").

- **Monthly Price (whole unit):**
    - Enter the total monthly rent for the apartment.

- **Special Instructions/Rules:**
    - Check this box if there are specific guidelines (e.g., "No pets allowed," "1-month advance, 1-month deposit").

- **Select Amenities:**
    - Tick applicable features such as Wi-Fi, Aircon, Parking, Laundry Area, Pet Friendly, or Furnished.
    - **Other Amenity:** Enter additional features not included in the listed options.
    - Tap Next to proceed to the location section.

2. Pin Apartment Location

- Opens a map interface for setting the apartment's exact location.
- Landlords can search the barangay or city using the search bar.
- **Pin Location on Map:** Drag the map to pin mark the apartment's location.
- The latitude and longitude coordinates automatically appear below the map.
- Tap Confirm to save the pinned location.

3. Submit Photos & 360° View

- **Thumbnail Photo (Optional):**
    - Upload a main photo that appears as the apartment's cover image.

- **Apartment Snippet Photos:**
    - Add multiple images showing various parts of the apartment (bedroom, kitchen, bathroom, etc.).

- **Video Tour (Optional):**
    - Upload a short walkthrough video (up to 2 minutes, MP4 format recommended).

- **360° View (Optional):**
    - Upload a panoramic image that allows tenants to look around the apartment in 360 degrees.
    - Tenants can pan, rotate the view, and use directional arrows to navigate different parts of the unit.
    - If no 360° image is uploaded, the apartment will simply display standard photos and videos.

- **Submit:**
    - Finalizes the listing upload.
    - The property is saved to the landlord's account and sent for admin verification before being visible to tenants.

III. Notifications

- A success message appears once the property is uploaded.
- The system notifies the landlord after verification and when the apartment becomes visible to tenants.
- Alerts appear if any required fields or media files are missing before submission.

MY PROPERTIES / MANAGE LISTINGS MODULE

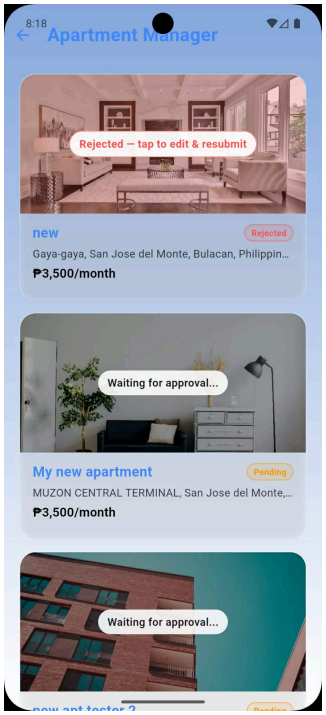


Figure 25. Property Management Page

I. Layout Overview

- Displays a list of all properties added by the landlord.
- Each property card shows the apartment's title, location, price, and current listing status (Active, Pending, or Inactive).
- Provides options to view, edit, or delete each property.

II. Functions

- **View Property:**
    - Opens the full apartment details as seen by tenants, including photos, amenities, and 360° view.
    - Allows the landlord to verify how the listing appears publicly.

- **Edit Property:**
    - Opens the property form pre-filled with existing information.

- Landlords can update details such as price, description, photos, or amenities.
- Changes are saved instantly and require admin re-verification if major edits are made.

- **Delete Property:**
  - Removes the listing permanently from the landlord's account and the tenant view.
  - A confirmation prompt appears before deletion to avoid accidental removals.

- **Listing Status:**
  - Displays whether the property is Active, Pending, or Inactive.
  - **Active:** Visible to tenants and open for booking.
  - **Pending:** Awaiting admin verification or re-approval after edits.
  - **Inactive:** Hidden from tenants but still stored in the landlord's account.

- **Search and Filter:**
  - Allows landlords to find specific properties by name, price, or status.

III. Notifications

- Sends alerts when a property is approved, edited, or deactivated.
- Notifies the landlord when tenants interact with a property (e.g., view, save, or book).

BOOKING REQUESTS MODULE

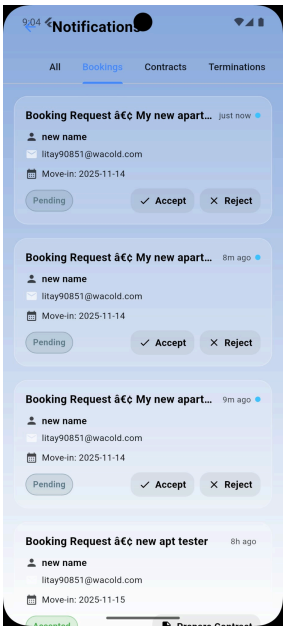


Figure 26. Booking Request Page

I. Layout Overview

- Displays all booking requests submitted by tenants for the landlord's listed properties.
- Each request card shows the tenant's name, selected apartment, preferred date, and booking status.
- Allows landlords to review, approve, or decline requests directly from the dashboard.

II. Functions

- **View Booking Details:**
    - Opens a detailed view of the booking request, showing tenant information, apartment name, requested schedule, and special notes.

- **Approve Booking:**
    - Confirms the tenant's booking request.

- Automatically updates the status to Approved and notifies the tenant instantly.
- Enables the tenant's Messaging Module for direct communication with the landlord.

- **Decline Booking:**
  - Rejects the booking request with an optional reason or note.
  - The tenant is notified that the booking was declined.

- **Pending Requests:**
  - Displays all requests awaiting landlord review.
  - Can be sorted by date or property name.

- **Approved Bookings:**
  - Shows confirmed bookings and upcoming tenant schedules.
  - Allows landlords to mark a booking as "Completed" after the rental period ends.

- **Declined Bookings:**
  - Lists all requests that were rejected, with timestamps and reasons.

- **Search and Filter:**
  - Landlords can filter booking requests by tenant name, property, or status (Pending, Approved, Declined).

III. Notifications

- Sends alerts to landlords for new incoming booking requests.
- Tenants receive notifications when a booking is approved or declined.
- The system updates booking statuses in real time across both landlord and tenant dashboards.

RENTAL CONTRACT MODULE

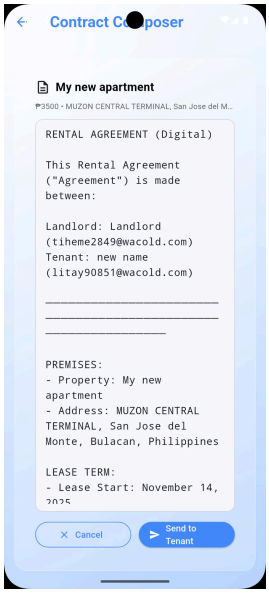


Figure 27. Contract Page

I. Layout Overview

- Displays the official rental agreement between the landlord and the tenant once a booking has been approved.
- Contains complete rental details including apartment name, rental price, duration, tenant information, and agreed house rules.
- Must be reviewed and agreed upon by both parties before proceeding to payment.

II. Functions

- **View Contract:**
    - Opens the rental contract generated automatically based on the approved booking details.
    - Displays key information such as rental terms, payment schedule, and both parties' names.

- **Edit Contract (Landlord):**
    - Allows the landlord to make minor adjustments such as adding additional clauses or updating rent amount before the tenant signs.

- Once finalized, the contract is locked and cannot be modified after both parties agree.

- **Tenant Agreement:**
  - The tenant can review the contract and agree to the terms by ticking the **"I Agree"** checkbox or signing electronically.
  - Payment options only become available once the tenant agrees.

- **Download Contract:**
  - Generates a PDF copy of the signed agreement for both the landlord and the tenant.
  - Saved copies can be accessed anytime under the contract history section.

- **Contract History:**
  - Displays all past and active rental agreements for the landlord's reference.
  - Includes details such as date signed, tenant name, and apartment name.

III. Notifications

- Both landlord and tenant receive confirmation once the contract is reviewed and signed.
- The system notifies the landlord when the tenant has completed their digital agreement.
- The **Payment Module** automatically activates after both parties confirm the contract.

PAYMENT VERIFICATION MODULE

I. Layout Overview

- Allows landlords to review and confirm rental payments submitted by tenants.
- Displays a list of recent payments with details such as tenant name, apartment name, payment method, amount, and status.
- Supports both **GCash (via PayMongo)** and **Cash Basis** payments.

II. Functions

- **View Payment Details:**
    - Opens the tenant's payment information including amount paid, date, and mode of payment.
    - Shows attached proof of payment (receipt or screenshot) if uploaded by the tenant.
- **Verify Payment:**
    - Confirms that the payment has been received and reviewed.
    - Updates the payment status to **Verified**, which notifies the tenant automatically.

- **Decline Payment:**
    - Marks a payment as invalid if the proof is unclear or the amount is incorrect.
    - The tenant is notified to re-upload proof or correct the payment.

- **Payment Methods:**
    - **GCash via PayMongo:** Displays the transaction reference number and confirms payment automatically once verified by PayMongo.
    - **Cash Basis:** Allows the landlord to manually verify the payment after confirming receipt in person.

- **View Payment History:**
    - Shows all previous payments, their verification status, and timestamps for reference.

- **Download Receipt:**
    - Generates a PDF receipt once a payment is successfully verified.
    - Both landlord and tenant can access the verified receipt under their respective accounts.

III. Notifications

- Landlords receive alerts for every new payment submitted by tenants.
- Tenants are notified immediately when their payment is approved or declined.
- The system updates payment status in real time across both landlord and tenant dashboards.

MAINTENANCE REQUESTS MODULE

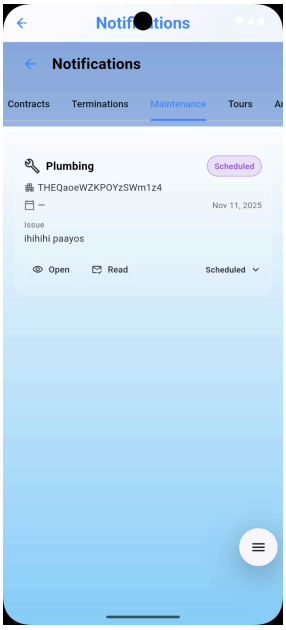


Figure 28. Maintenance Request Page

I. Layout Overview

- Allows landlords to view, manage, and respond to maintenance or repair requests submitted by tenants.
- Displays all requests in a list format, showing the tenant's name, property, issue category, and request status.
- Helps ensure timely repairs and proper communication between landlord and tenant.

II. Functions

- **View Request Details:**
    - Opens the full maintenance report submitted by the tenant, including the issue description, date, and attached photo or video.

- **Update Status:**
    - Landlords can change the progress of a request to Pending, In Progress, or Completed.

- Each status update notifies the tenant automatically.

- **Assign Maintenance Personnel:**
  - Landlords can assign the issue to a maintenance staff member or technician.
  - The assigned personnel details appear in the tenant's view for transparency.

- **Attach Response:**
  - Allows landlords to send a short reply or update regarding the issue.

- **View Completed Requests:**
  - Displays all resolved maintenance reports for record-keeping and reference.

- **Search and Filter:**
  - Enables filtering of requests by property, date, or status.

III. Notifications

- Sends alerts to landlords when a new maintenance request is submitted
- Tenants receive notifications when their request is updated or marked as completed.
- The system keeps a history of all maintenance activities for future reference.

MESSAGING MODULE



Figure 29. Messaging Page

I. Layout Overview

- Provides a built-in chat system for direct communication between landlords and tenants.
- Accessible only when a tenant has an approved booking or active rental agreement.
- Displays a list of ongoing conversations, each labeled with the tenant's name and apartment title.

II. Functions

- **Open Chat:**
    - Opens a conversation thread between the landlord and a specific tenant.
    - Shows previous messages, attachments, and timestamps for easy reference.

- **Send Message:**
    - Allows landlords to send text messages or quick updates to the tenant.
    - Messages appear instantly in the tenant's chat interface.

- **Attach File:**
    - Enables landlords to send documents, images, or proof related to the rental or maintenance.

- **View Conversation History:**
    - Displays all past messages with each tenant, organized chronologically.
    - Older messages remain accessible for record and reference.

- **End Chat / Archive Conversation:**
    - Available once the rental period is completed.
    - Closes or archives the chat thread but keeps a copy in the message history for record-keeping.

III. Notifications

- Landlords receive real-time notifications when a tenant sends a message.
- Unread messages are marked with a badge icon on the Messages tab.
- Tenants are notified instantly when landlords reply.

LANDLORD PROFILE MODULE

I. Layout Overview

- Displays the landlord's personal and business information such as name, contact number, email, and profile photo.
- Includes options to edit personal details, update profile photo, and view verification information.

II. Functions

- **Edit Profile:**
    - Allows landlords to update their personal information, contact number, or email address.
    - Saves all changes automatically once updated.

- **Change Profile Photo:**
    - Enables landlords to upload or replace their display image.
    - The updated photo appears on their profile and dashboard.

- **Account Settings:**
    - Provides options to change password, manage security preferences, or enable two-factor authentication (if available).

- **Logout:**
    - Safely logs the landlord out of the system and redirects to the Login screen.

III. Notifications

- Displays confirmation messages when profile updates are successfully saved.
- Sends alerts if verification requires renewal or additional documents.

HELP & SUPPORT MODULE

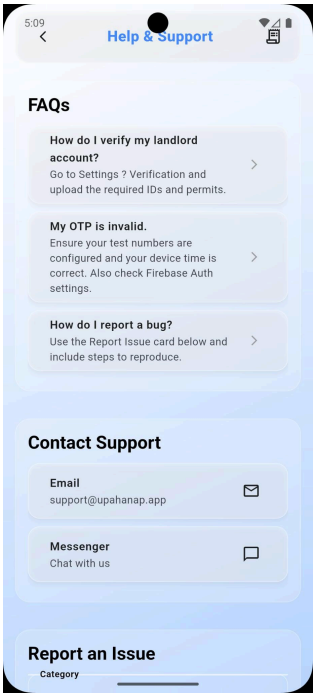


Figure 30. Help & Support Page

I. Layout Overview

- Provides landlords with a direct way to contact the UpaHanap admin team for assistance or inquiries.
- Includes options to submit a support ticket, attach files, and view previous messages with the admin.
- Ensures that all concerns are properly tracked and resolved within the system.

II. Functions

- **New Support Request:**
    - Opens a form where landlords can describe their issue, question, or concern.

- Can include topics such as listing problems, payment issues, or account verification.

- **Category Selection:**
  - Allows landlords to classify the concern (e.g., Technical Issue, Payment, Account, Other).

- **Attach File:**
  - Enables uploading of screenshots, receipts, or documents to support the inquiry.

- **Send Message:**
  - Submits the support request to the Super Admin or technical team for review.
- **View Replies:**
  - Displays admin responses or updates in the same conversation thread.

- **Close Ticket:**
  - Marks a resolved issue as completed and moves it to the closed tickets list.

III. Notifications

- Sends alerts when the admin responds to a support request.
- Updates the ticket status in real time (Pending, In Progress, Resolved).
- Notifies the landlord once a concern is officially closed.

APPENDIX C. TESTING INTERFACE

Test Case 1. Registration

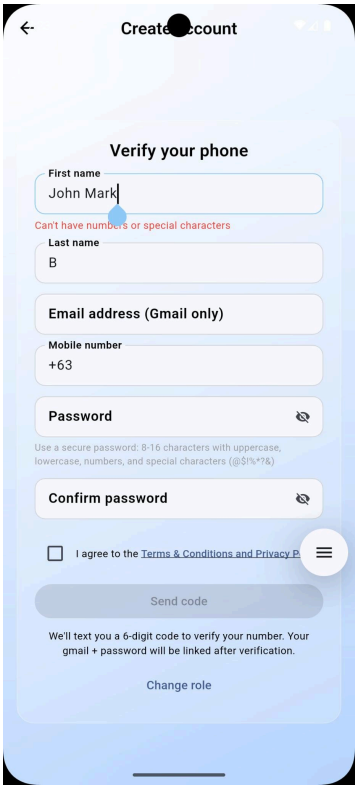


Figure 31. Registration Test Case 1.1

Case 1.1:

- First Name: The user trying to input a name containing a number or special character.

  Result: The system shows “Can’t have numbers or special characters.
- All fields are not filled
- *Result:* The “Send Code” button is disabled.

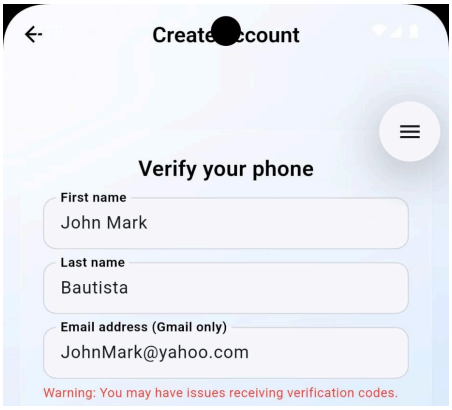


Figure 31.2. Registration Test Case 1.2

Case 1.2:

- Email: The user inputs a Yahoo address instead of Gmail.
  Result: A warning appears saying "You may have issues receiving verification codes." The "Send Code" button is disabled.

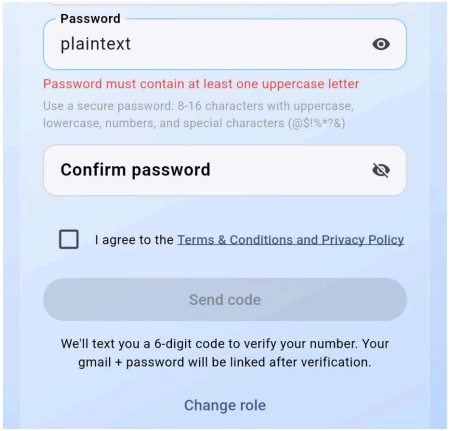


Figure 31.3. Registration Test Case 1.3

Case 1.3:

- Password: The user inputs a password without an uppercase letter.
  Result: The system shows "Password must contain at least one uppercase letter."

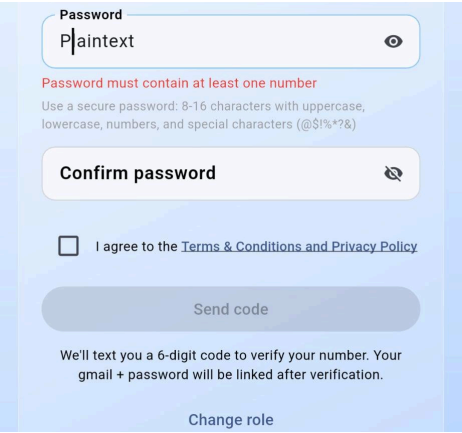


Figure 31.4. Registration Test Case 1.4

Case 1.4:

- Password: The user inputs a password without a number.

  Result: The system shows "Password must contain at least one number."

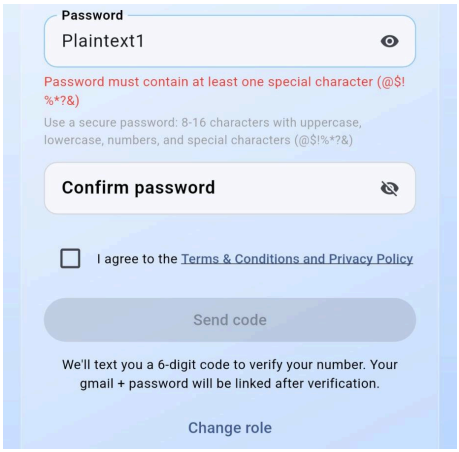


Figure 31.5. Registration Test Case 1.5

Case 1.5:

- Password: The user inputs a password without a special character.

  Result: The system shows "Password must contain at least one special character (@$!%*?&)."

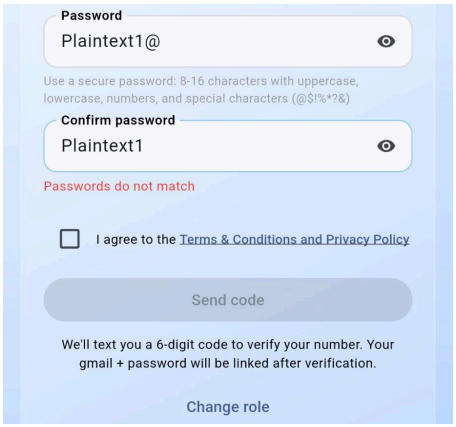


Figure 31.6. Registration Test Case 1.6

Case 1.6:

- Password: Plaintext1@
- Confirm Password: Plaintext1

Result: The system shows "Passwords do not match."

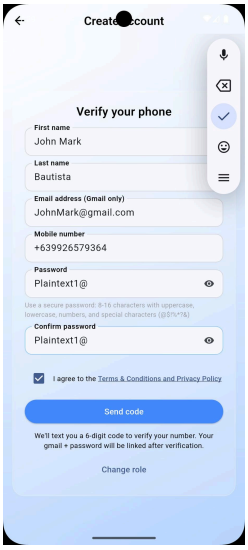


Figure 31.7. Registration Test Case 1.7

Case 1.7:

- Email: The user inputs a valid Gmail address.
- Password: The user inputs a strong password.
- Confirm Password: Matches the password.

- Terms & Conditions and Privacy Policy: The user checks the agreement box.
  Result: The system accepts all inputs, the *Send Code* button is enabled, and the user can proceed to receive the verification code.

Test Case 2. Add Property and Submit Photos

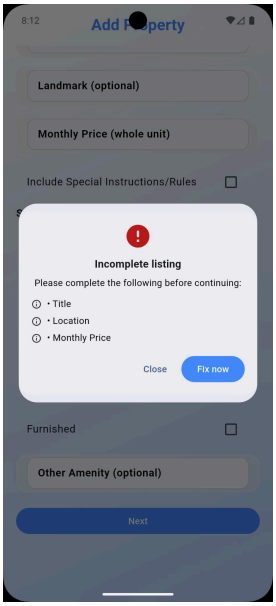


Figure 32.1. Add Property Test Case 2.1

Case 2.1:

- The user leaves the Title, Location, and Monthly Price fields empty.
  Result: The system displays an *"Incomplete Listing"* dialog showing missing fields (Title, Location, Monthly Price).

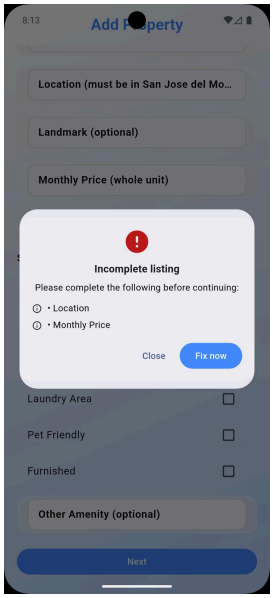


Figure 32.2. Add Property Test Case 2.2

Case 2.2:

- Title: The user fills in *"My new apartment."*
- Location: The user leaves the field blank.
- Monthly Price: The user leaves the field blank.

  Result: The system displays an *"Incomplete Listing"* pop-up message saying:

  "Please complete the following before continuing:

  • Location

  • Monthly Price."

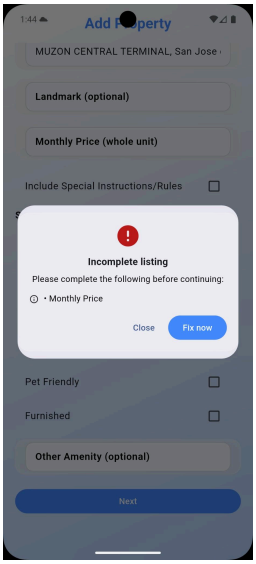


Figure 32.3. Add Property Test Case 2.3

Case 2.3:

- Title: Entered correctly.
- Location: Filled correctly.
- Monthly Price: Left blank.

  Result: The system displays an error dialog *"Please complete Monthly Price before continuing."*

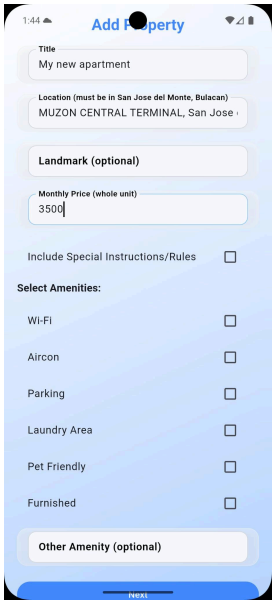


Figure 32.4. Add Property Test Case 2.4

Case 2.4:

- Title: The user inputs *"My new apartment."*
- Location: The user selects *"Muzon Central Terminal."*
- Monthly Price: The user inputs *₱3,500.*
- Amenities: The user selects *Wi-Fi, Aircon, Parking.*

  Result: The system accepts the inputs. User clicks the Next button, and proceeds to the photo submission page.

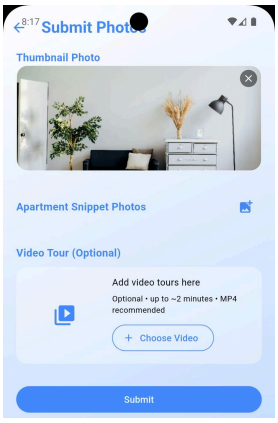


Figure 32.5. Submit Photos Test Case 2.5

Case 2.5:

- The user uploads one Thumbnail Photo.
- The user does not upload any snippet photos or videos.

  Result: The system accepts the thumbnail photo.

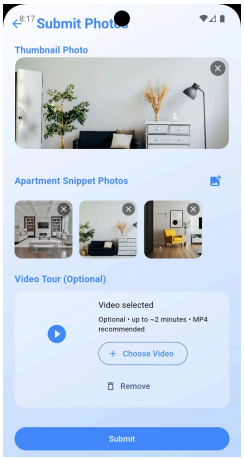


Figure 32.6. Submit Photos Test Case 2.6

Case 2.6:

- The user uploads 1 Thumbnail Photo, 3 Snippet Photos, and an optional Video Tour (≤2 mins).

  Result: The system accepts all uploads, shows thumbnails with a “Remove” option.

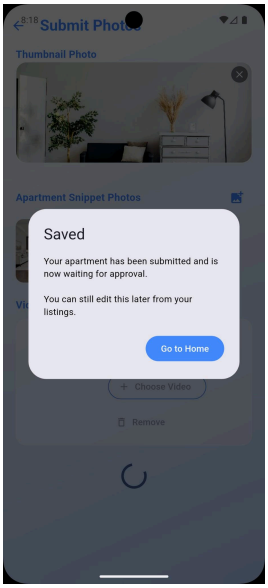


Figure 32.7. Submit Photos Test Case 2.7

Case 2.7:

- The user uploads all required photos (thumbnail and snippet).
- The user clicks Submit.

Result: A confirmation message appears saying:
"Your apartment has been submitted and is now waiting for approval. You can still edit this later from your listings."
The system saves the apartment details and redirects the user to Home after clicking *Go to Home*.

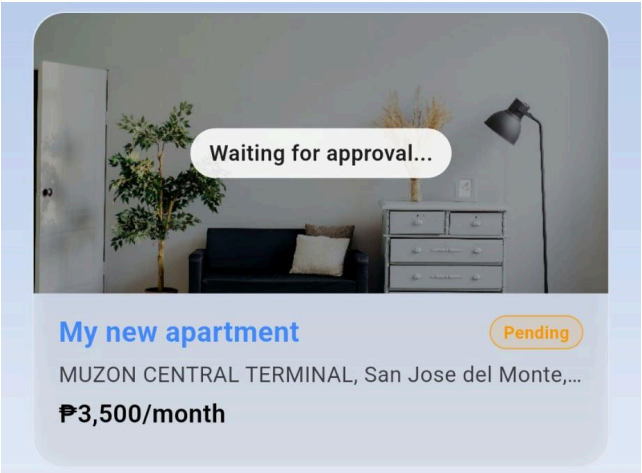


Figure 32.6. Apartment Manager Test Case 2.6

Case 2.6:

- The user submits the listing.

  Result: The new apartment appears in the Apartment Manager screen with status "Pending – Waiting for Approval."

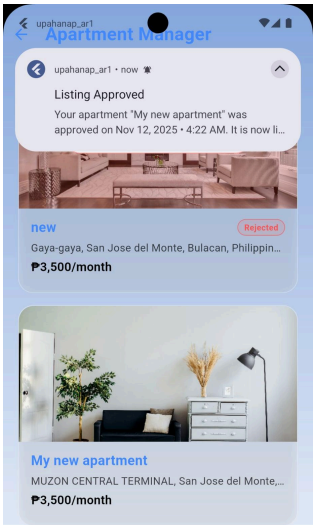


Figure 32.8. Apartment Manager Test Case 2.8

Case 2.8:

- The submitted apartment is pending approval by the admin.

Result: Once reviewed and approved, a notification appears saying:

"Listing Approved — Your apartment 'My new apartment' was approved on [date and time]."

The status of the listing changes from Pending to Approved in the *Apartment Manager* screen.

The status of the specific apartment card within the *Apartment Manager* also updates automatically to Approved, reflecting its current state.

Test Case 3. Tenant Schedule a Tour Process

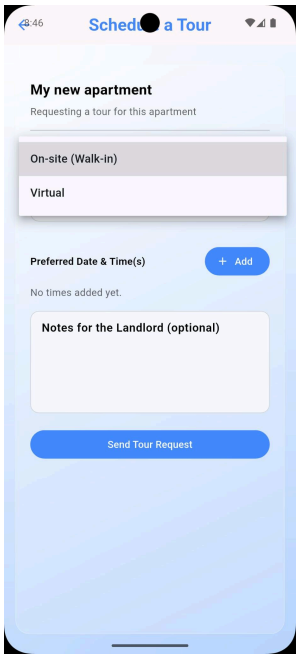


Figure 33.1. Choosing Tour Type Test Case 3.1

Case 3.1:

- The tenant opens the *Schedule a Tour* page for My new apartment.

- The tenant selects a tour type from the dropdown (e.g., *On-site (Walk-in)* or *Virtual*).

  Result: The chosen tour type is displayed in the Tour Type field and can be changed anytime before submission.

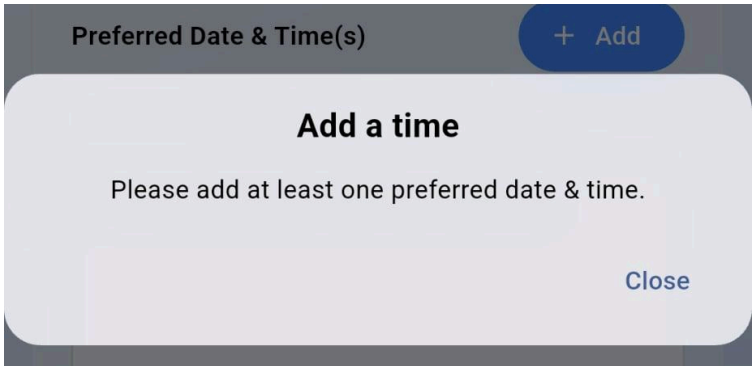


Figure 33.3. Missing Date and Time Test Case 3.3

Case 3.2:

- The tenant attempts to send the tour request without adding any preferred date or time.

  Result: A pop-up alert appears.

  The form remains open for correction.

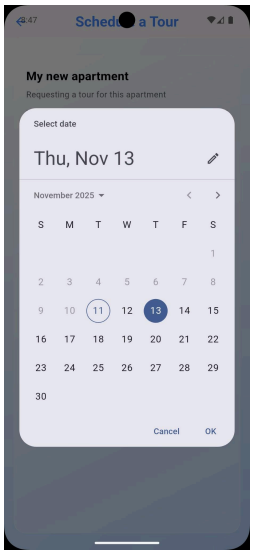


Figure 33.3. Adding Preferred Date and Time Test Case 3.3

Case 3.3:

- The tenant taps the Add button under *Preferred Date & Time(s)*.

- The tenant selects a date from the calendar (e.g., November 13, 2025) and confirms it.

  Result: The selected date is added to the *Preferred Date & Time(s)* list, ready for submission.

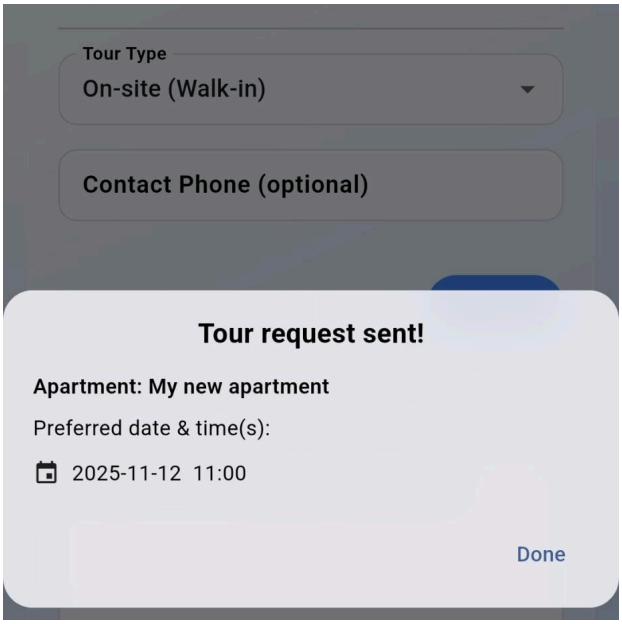


Figure 33.4. Submitting Tour Request Test Case 3.4

Case 3.4:

- The tenant fills in all required fields.

- The tenant taps Send Tour Request to confirm the tour.

  Result: A confirmation pop-up appears saying:

  "Tour request sent!

  Apartment: My new apartment

  Preferred date & time(s): 2025-11-12 11:00"

  The tenant can tap Done to close the dialog.

Test Case 4. Tenant Booking and Lease Process

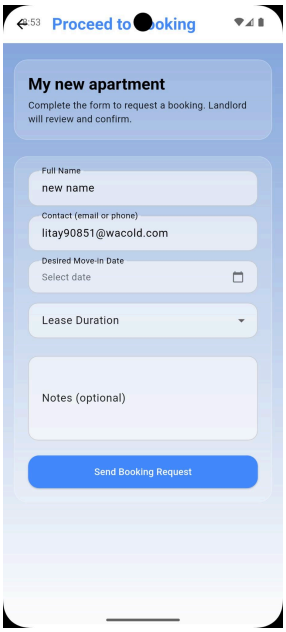


Figure 34.1. Booking Page Test Case 4.1

Case 4.1:

- The tenant opens the booking page for *"My new apartment."*
  Result: The fields Full Name and Contact are automatically filled based on the tenant's profile.

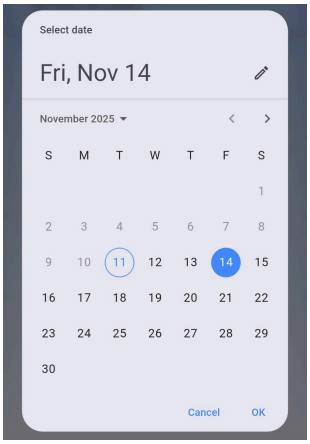


Figure 34.2. Select Move-in Date Test Case 4.2

Case 4.2:

- The tenant taps the calendar icon and selects November 14, 2025 as the desired move-in date.

  Result: The date picker closes and the selected date appears in the field as.

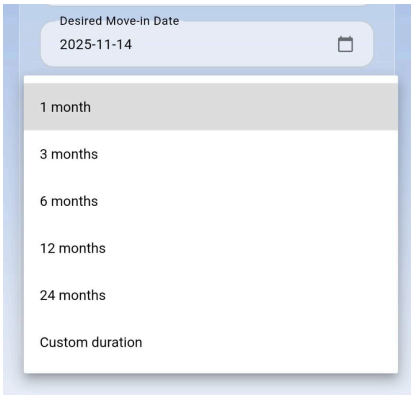


Figure 34.3. Lease Duration Test Case 4.3

Case 4.3:

- The tenant opens the Lease Duration dropdown and selects 12 months.

  Result: The lease duration field displays 12 months.

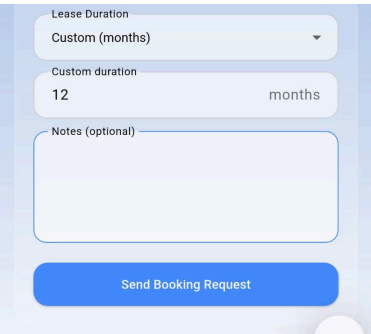


Figure 34.4. Custom Lease Duration Test Case 4.4

Case 4.4:

- The tenant selects Custom Duration from the list.
- The tenant inputs 12 in the months field.

  Result: The field accepts the input and displays 12 months as the custom lease duration.

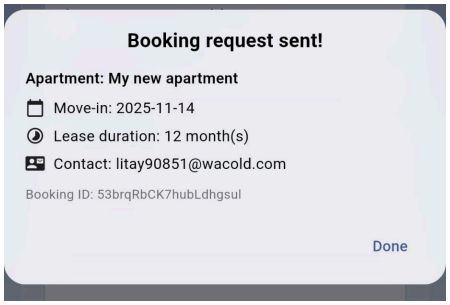


Figure 34.5. Booking Request Test Case 4.5

Case 4.5:

- The tenant completes all required fields.
- The tenant taps Send Booking Request.

  Result: A pop-up message appears.

  The tenant can tap Done to close the dialog.

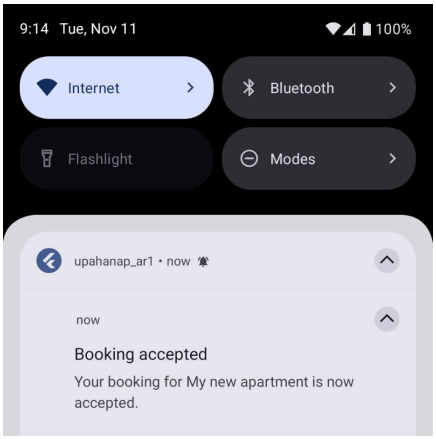


Figure 34.6. Booking Accepted Notification Test Case 4.6

Case 4.6:

- The landlord approves the booking request.
- The tenant receives a system notification.

  Result: A notification appears stating:

  "Booking Accepted — Your booking for My new apartment is now accepted."

  It is also reflected on the app's notification page.

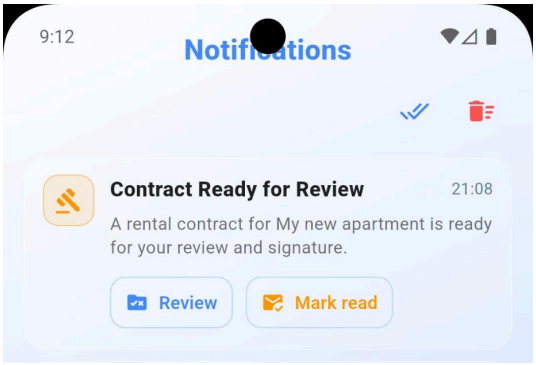


Figure 34.7. Contract Ready for Review Test Case 4.7

Case 4.7:

- After the booking is accepted, the system generates a digital rental contract.

- The tenant receives a notification to review the contract.

  Result: A notification appears saying:

"Contract Ready for Review — A rental contract for My new apartment is ready for your review and signature."

The tenant can open and view full contract details such as rent, deposit, landlord, and tenant information.

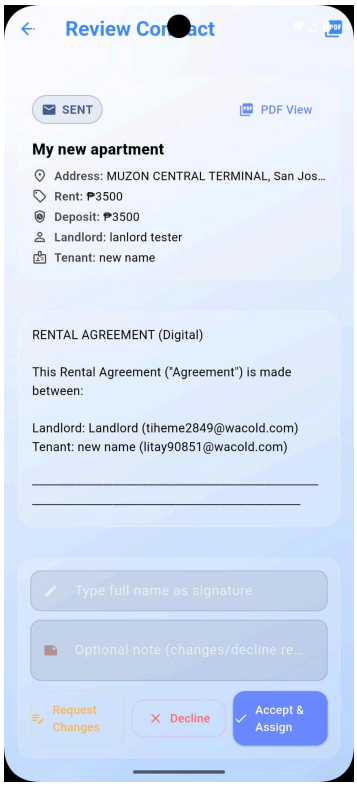


Figure 34.8. Contract Acceptance Test Case 4.8

Case 4.8:

- The tenant opens the contract page.
- The tenant reviews the contract details.
- The tenant taps Accept & Assign to finalize the agreement.

  Result: The system finalizes the contract and confirms the lease. The status updates to Tenant Accepted Lease, and the apartment details appear in the tenant's dashboard.

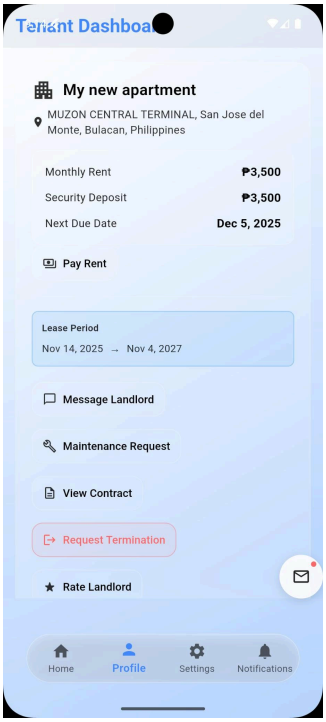


Figure 34.9. Tenant Dashboard Test Case 4.9

Case 4.9:

- The tenant opens their dashboard after signing the contract.
- The system displays the newly accepted lease.

  Result: The apartment appears under My new apartment showing:
- Address: Muzon Central Terminal, San Jose del Monte, Bulacan
- Monthly Rent: ₱3,500
- Security Deposit: ₱3,500
- Next Due Date: Dec 5, 2025
- Lease Period: Nov 14, 2025 → Nov 4, 2027

  Additional actions like Message Landlord, View Contract, Request Termination, and Rate Landlord are available.

APPENDIX D. SOURCE CODE



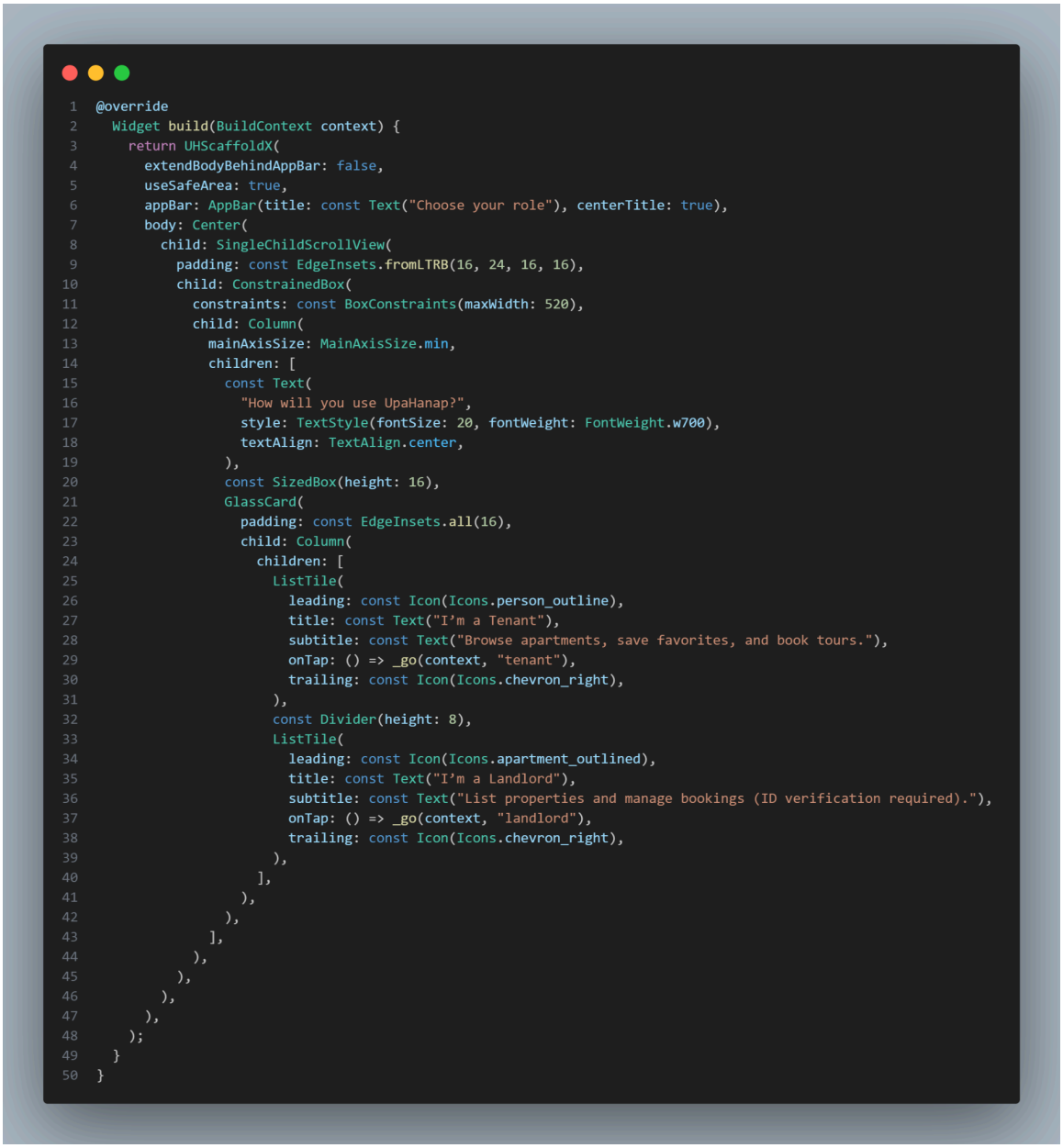
```
    location: widget.location,
                price: widget.price,
                imageUrl: widget.imageUrl,
                landmark: widget.landmark,
                landlord: widget.landlord,
                isAvailable: widget.isAvailable,
                isFurnished: widget.isFurnished,
                amenities: widget.amenities,
                specialInstructions: widget.specialInstructions,
                otherAmenity: widget.otherAmenity,
              ),
              scrollable: true,
            ),
            _buildGlassCard(
              LandlordPhotosScreen(images: _gallery),
              scrollable: false,
            ),
            _buildGlassCard(
              _loadingVideos
                  ? const Center(child: CircularProgressIndicator())
                  : LandlordVideoToursScreen(videos: _effectiveVideos),
              scrollable: false,
            ),
            _buildGlassCard(
              ARViewScreen(isAvailable: widget.isAvailable),
              scrollable: true,
            ),
          ],
        ),
      ),

      // map prompt
      GestureDetector(
        onVerticalDragUpdate: (details) {
          if (details.primaryDelta != null && details.primaryDelta! < -6) {
            _openMapSheet();
          }
        },
        onTap: _openMapSheet,
        child: const Column(
          children: [
            Icon(Icons.keyboard_arrow_up, color: Colors.blueAccent, size: 28),
            Text(
              "Swipe up to view in a map",
              style: TextStyle(color: Colors.blueAccent, fontSize: 14),
            ),
            SizedBox(height: 8),
          ],
        ),
      ),

      SmoothPageIndicator(
        controller: _pageController,
        count: 4,
        effect: const ExpandingDotsEffect(
          activeDotColor: Colors.blueAccent,
          dotColor: Colors.white24,
          dotHeight: 10,
          dotWidth: 10,
        ),
      ),
      SafeArea(
    child: Padding(
      padding: const EdgeInsets.symmetric(horizontal: 20, vertical: 12),
      child: Row(
        children: [
          // Schedule a Tour
          Expanded(
            child: OutlinedButton(
              onPressed: () {
                final user = FirebaseAuth.instance.currentUser;
                if (user == null) {
                  _showLoginRequiredPopup();
                } else {
                  Navigator.push(
                    context,
                    MaterialPageRoute(
                      builder: (_) => TourScheduleScreen(
                        apartmentId: widget.apartmentId,
                        apartmentTitle: widget.title,
                        landlordId: widget.landlordId,
                      ),
                    ),
                  );
                }
              },
              child: const Text("Schedule a Tour"),
            ),
          ),
          const SizedBox(width: 12),
          // Proceed to Booking
          Expanded(
            child: FilledButton(
              onPressed: () {
                final user = FirebaseAuth.instance.currentUser;
                if (user == null) {
                  _showLoginRequiredPopup();
                } else {
                  Navigator.push(
                    context,
                    MaterialPageRoute(
                      builder: (_) => NewBookingFormScreen(
                        apartmentId: widget.apartmentId,
                        apartmentTitle: widget.title,
                        landlordId: widget.landlordId,
                        availableDates: widget.availableDates,
                      ),
                    ),
                  );
                }
              },
              child: const Text("Proceed to Booking"),
            ),
          ),
        ],
      ),
    ),
  )


    ],
  );
}
```

Figure 35. Booking UI Code



```
@override
  Widget build(BuildContext context) {
    return UHScaffoldX(
      extendBodyBehindAppBar: false,
      useSafeArea: true,
      appBar: AppBar(title: const Text("Choose your role"), centerTitle: true),
      body: Center(
        child: SingleChildScrollView(
          padding: const EdgeInsets.fromLTRB(16, 24, 16, 16),
          child: ConstrainedBox(
            constraints: const BoxConstraints(maxWidth: 520),
            child: Column(
              mainAxisSize: MainAxisSize.min,
              children: [
                const Text(
                  "How will you use UpaHanap?",
                  style: TextStyle(fontSize: 20, fontWeight: FontWeight.w700),
                  textAlign: TextAlign.center,
                ),
                const SizedBox(height: 16),
                GlassCard(
                  padding: const EdgeInsets.all(16),
                  child: Column(
                    children: [
                      ListTile(
                        leading: const Icon(Icons.person_outline),
                        title: const Text("I'm a Tenant"),
                        subtitle: const Text("Browse apartments, save favorites, and book tours."),
                        onTap: () => _go(context, "tenant"),
                        trailing: const Icon(Icons.chevron_right),
                      ),
                      const Divider(height: 8),
                      ListTile(
                        leading: const Icon(Icons.apartment_outlined),
                        title: const Text("I'm a Landlord"),
                        subtitle: const Text("List properties and manage bookings (ID verification required)."),
                        onTap: () => _go(context, "landlord"),
                        trailing: const Icon(Icons.chevron_right),
                      ),
                    ],
                  ),
                ),
              ],
            ),
          ),
        ),
      ),
    );
  }
}
```

Figure 36. Role Picker Code



```
// Lease Duration (with custom months)
DropdownButtonFormField<String>(
  value: _leaseDuration,
  onChanged: (v) {
    setState(() {
      _leaseDuration = v;
      _customMonthsError = null;
    });
    if (v == 'Custom (months)') {
      Future.delayed(
          const Duration(milliseconds: 100), () {
        if (mounted)
          _customMonthsFocus.requestFocus();
      });
    }
  },
  items: _leaseDurations
      .map((d) =>
          DropdownMenuItem(value: d, child: Text(d)))
      .toList(),
  style: const TextStyle(color: Colors.black),
  decoration: _inputStyle(label: 'Lease Duration'),
  dropdownColor: Colors.white,
  validator: (v) => (v == null || v.isEmpty)
      ? 'Please select a lease duration'
      : null,
),
const SizedBox(height: 12),

// Custom months numeric field
AnimatedSwitcher(
  duration: const Duration(milliseconds: 200),
  switchInCurve: Curves.easeOut,
  switchOutCurve: Curves.easeIn,
  child: _isCustom
      ? TextFormField(
          key: const ValueKey('customMonthsField'),
          controller: _customMonthsCtrl,
          focusNode: _customMonthsFocus,
          keyboardType: TextInputType.number,
          inputFormatters: [
            FilteringTextInputFormatter.digitsOnly,
          ],
          onChanged: (val) {
            // live warning if letters found (should be filtered anyway)
```

Figure 37. Booking Code



```
return Stack(
            children: [
              SafeArea(
                child: Padding(
                  padding: const EdgeInsets.all(16),
                  child: GlassCard(
                    blurSigma: 12,
                    tintOpacity: 0.25,
                    borderRadius: BorderRadius.circular(20),
                    child: Padding(
                      padding: const EdgeInsets.all(16),
                      child: Column(
                        children: [
                          Row(
                            children: [
                              Icon(
                                _isEditMode
                                    ? Icons.edit_document
                                    : Icons.description_outlined,
                              ),
                              const SizedBox(width: 8),
                              Expanded(
                                child: Text(
                                  _resolvedPropertyTitle.isNotEmpty
                                      ? _resolvedPropertyTitle
                                      : (_booking['propertyTitle'] ??
                                          'Rental Contract'),
                                  style: const TextStyle(
                                    fontSize: 18,
                                    fontWeight: FontWeight.w700,
                                  ),
                                ),
                              ),
                              const SizedBox(width: 8),
                              IconButton(
                                tooltip: 'Preview',
                                icon: const Icon(Icons.remove_red_eye_outlined),
                                onPressed: _openPreview,
                              ),
                            ],
                          ),
                          const SizedBox(height: 8),
                          Align(
                            alignment: Alignment.centerLeft,
                            child: Text(
                              '₱${_resolvedMonthlyRent.toString()} • $_resolvedPropertyAddress',
                              maxLines: 1,
                              overflow: TextOverflow.ellipsis,
                              style: TextStyle(
                                fontSize: 12,
                                color: Colors.grey.shade700,
                              ),
                            ),
                          ),
                          const SizedBox(height: 12),
                          Expanded(
                            child: TextField(
                              controller: _controller,
                              maxLines: null,
                              decoration: const InputDecoration(
                                border: InputBorder.none,
                                hintText: 'Contract text…',
                              ),
                              style: const TextStyle(fontFamily: 'monospace'),
                            ),
                          ),
                          const SizedBox(height: 12),
                          Row(
                            children: [
                              Expanded(
                                child: OutlinedButton.icon(
                                  onPressed: _saving
                                      ? null
                                      : () => Navigator.pop(context),
                                  icon: const Icon(Icons.close),
                                  label: const Text('Cancel'),
                                ),
                              ),
                              const SizedBox(width: 12),
                              Expanded(
                                child: ElevatedButton.icon(
                                  onPressed: _saving ? null : _sendToTenant,
                                  icon: _saving
                                      ? const SizedBox(
                                          height: 16,
                                          width: 16,
                                          child: CircularProgressIndicator(
                                            strokeWidth: 2,
                                          ),
                                        )
                                      : const Icon(Icons.send),
                                  label: Text(
                                    _saving
                                        ? 'Sending…'
                                        : _isEditMode
                                            ? 'Update & Send'
                                            : 'Send to Tenant',
                                  ),
                                ),
                              ),
                            ],
                          ),
                        ],
                      ),
                    ),
                  ),
                ),
              ),
            ],
          );
```

Figure 38. Contract Sender to Tenants



```
SizedBox(
                    width: double.infinity,
                    child: ElevatedButton(
                      onPressed: _busy || !_agreed ? null : _sendOtp,
                      child: _busy
                          ? const SizedBox(
                              width: 18,
                              height: 18,
                              child: CircularProgressIndicator(
                                strokeWidth: 2,
                                color: Colors.white,
                              ),
                            )
                          : const Text("Send code"),
                    ),
                  ),
                  const SizedBox(height: 8),
                  const Text(
                    "We'll text you a 6-digit code to verify your number. Your gmail + password will be linked after verification.",
                    style: TextStyle(fontSize: 12),
                    textAlign: TextAlign.center,
                  ),
```

Figure 39. Otp Sender



```
Material(
                  color: Colors.white, // ✅ solid white
                  borderRadius: BorderRadius.circular(20),
                  elevation: 10, // nice lift
                  shadowColor: Colors.black26,
                  clipBehavior: Clip.antiAlias, // keep rounded corners
                  child: SingleChildScrollView(
                    controller: scrollController,
                    padding: const EdgeInsets.fromLTRB(16, 16, 16, 80),
                    child: Column(
                      crossAxisAlignment: CrossAxisAlignment.start,
                      children: [
                        Text(
                          apartment.title,
                          style: Theme.of(context)
                              .textTheme
                              .titleLarge
                              ?.copyWith(fontWeight: FontWeight.bold),
                        ),
                        const SizedBox(height: 6),
                        Text(
                          apartment.location,
                          style: Theme.of(context)
                              .textTheme
                              .bodyMedium
                              ?.copyWith(color: Colors.grey[700]),
                        ),
                        const SizedBox(height: 8),
                        Text("Owner: ${apartment.landlord}"),
                        const SizedBox(height: 8),
                        Text("Landmark: ${apartment.landmark}"),
                        const SizedBox(height: 8),
                        Text(
                            "Furnished: ${apartment.isFurnished ? 'Yes' : 'No'}"),
                        Text(
                            "Available: ${apartment.isAvailable ? 'Yes' : 'No'}"),
                        const SizedBox(height: 8),
                        if (apartment.amenities.isNotEmpty)
                          Text("Amenities: ${apartment.amenities.join(', ')}"),
                        const SizedBox(height: 14),
                        Text(
                          "Php ${apartment.price.toStringAsFixed(2)}/month",
                          style: Theme.of(context)
                              .textTheme
                              .titleMedium
                              ?.copyWith(fontWeight: FontWeight.w700),
                        ),
```

Figure 40. Apartment Details Code

APPENDIX E. CURRICULUM VITAE OF DEVELOPER

Curriculum Vitae of

John Mark D.C. Bautista

Block 11, Lot 7, Altamonte Street, Marilao Green Subdivision, Loma de Gato Marilao, Bulacan
bautistajohnmark49@gmail.com
09687048086

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	June 2022 - present	STI College San Jose del Monte
High School	June 2016 - 2022	Prenza National High School
Elementary	June 2011 - 2016	Loma de Gato Elementary School

SKILLS

SKILLS	Level of Competency	Date Acquired
Java	Intermediate	September 2022
C#	Intermediate	October 2023
Python	Intermediate	February 2024

Curriculum Vitae of

Jomari R. Pista

Del Valle, Daang Barrio Road, Brgy. Poblacion I, City of San Jose del Monte, Bulacan
pistajomari@gmail.com
09946567898

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary Senior High School	June 2022 - present June 2020-2022	STI College San Jose del Monte San Jose del Monte National High School
Junior High School	June 2012 - 2016	San Jose del Monte National High School
Elementary	June 2011 - 2016	San Jose del Monte Central Elementary School

SKILLS

SKILLS	Level of Competency	Date Acquired
Java	Intermediate	September 2022
C#	Intermediate	October 2023
Python	Intermediate	February 2024

Curriculum Vitae of

Nickos B. Poblete

Block 3, Lot 14, Heoresville 3 Housing Project, Gaya-Gaya, San Jose del Monte, Bulacan
pobletenickos5@gmail.com
09948227592

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	June 2022 - present	STI College San Jose del Monte
High School	June 2016 - March 2022	St. Raphael College of business and Arts Inc.
Elementary	June 2011 - March 2016	Kiloloron Elementary School

SKILLS

SKILLS	Level of Competency	Date Acquired
Java	Intermediate	September 2022
Python	Intermediate	January 2023
C#	Intermediate	February 2024

Curriculum Vitae of

Denver Mavric M. Rabino

Block 7 Lot 27 Phase 8A Bagong Silang Caloocan City
denvermavvricrabino@gmail.com
09947890045

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary Tertiary	August 2023-present August 2021-2023	STI College San Jose del Monte FEU Institute of Technology
High School	June 2014 - 2018	Holy Angel School of Caloocan Inc.
Elementary	June 2008 - 2014	Saint James Academy

SKILLS

SKILLS	Level of Competency	Date Acquired
C++ Java	Intermediate Intermediate	August 2021 September 2022
C#	Intermediate	October 2023
Python	Intermediate	February 2024